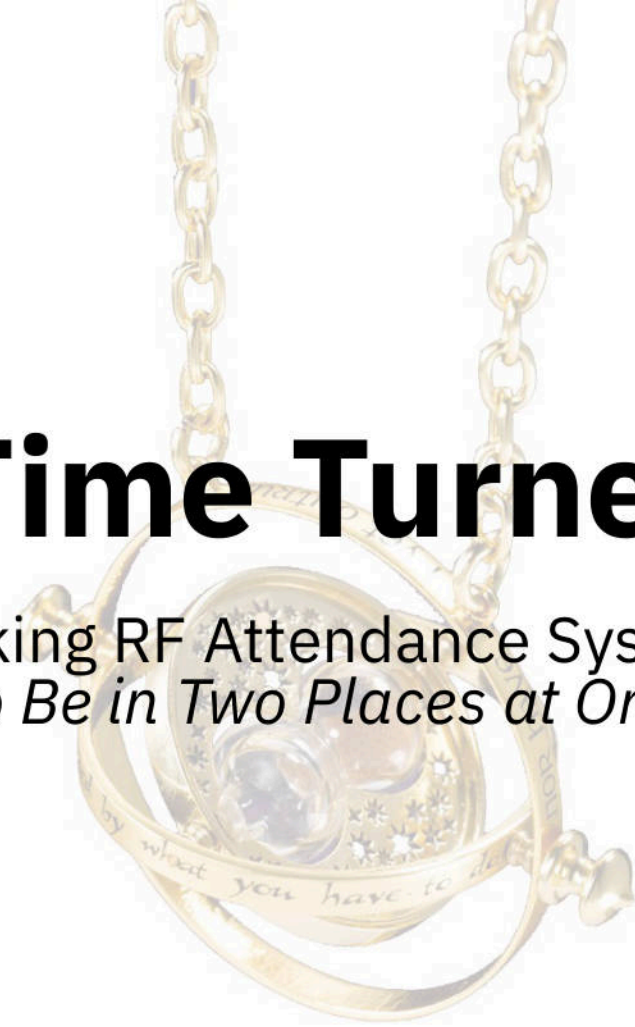




Time Turner

Hacking RF Attendance Systems
(To Be in Two Places at Once)





✓ 1) ORIENTATION AND PROFESSIONAL DEVELOPMENT

FA18	ENG 100 CS8	0.0	S
FA19	ECE 316 E2	3.0	A+

✓ 2) CORE MATHEMATICS COURSES

FA18	MATH 220 1	4.0	PS
FA18	MATH 241 CL2	4.0	A-
GS19	MATH 415 M16	3.0	A
GS20	MATH 231 AD2	3.0	A

✓ 3) PHYSICS SEQUENCE

SP20	PHYS 212 A11	4.0	A+
FA20	PHYS 211 A2	4.0	A+

✓ 4) SCIENCE ELECTIVE
(COURSE LIST ON DEPARTMENT WEBSITE)

FA18	FSHN 120 A	3.0	A
------	------------	-----	---

✗ 5) COMPUTER SCIENCE REQUIREMENTS

NEEDS: 2 COURSES

SELECT FROM: CS 374, CS 421

1) ORIENTATION AND PROFESSIONAL DEVELOPMENT

FA18	ENG 100 CS8	0.0	S
FA19	ECE 316 E2	3.0	A+

2) CORE MATHEMATICS COURSES

FA18	MATH 220 1	4.0	PS
FA18	MATH 241 CL2	4.0	A-
GS19	MATH 415 M16	3.0	A
GS20	MATH 231 AD2	3.0	A

3) PHYSICS SEQUENCE

SP20	PHYS 212 A11	4.0	A+
FA20	PHYS 211 A2	4.0	A+

4) SCIENCE ELECTIVE (COURSE LIST ON DEPARTMENT WEBSITE)

FA18	FSHN 120 A	3.0	A
------	------------	-----	---

5) COMPUTER SCIENCE REQUIREMENTS

NEEDS: 2 COURSES

SELECT FROM: CS 374, CS 421

Schedule

Schedule Details

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
00							
01							
02		CS374-ADC 02:50-03:13 (Pending)	CS421-D4 02:15-03:10 (Pending)				
03							
04							
05							

Panels

Summary

Title	Details	Hour	CRN	Schedule Typ	Status	Action
Intro to Algs & Mode...	CS 374, ADC	0	70645	Discussion...	Pending	**Web Registered**
Progrmg Languages...	CS 421, D4	4	40087	Lecture-Di...	Pending	**Web Registered**

Total Hours | Registered: 0 | Billing: 0 | CEU: 0 | Min: 0 | Max: 0

Submit

1) ORIENTATION AND PROFESSIONAL DEVELOPMENT

FA18	ENG 100 CS8	0.0	S
FA19	ECE 316 E2	3.0	A+

2) CORE MATHEMATICS COURSES

FA18	MATH 220 1	4.0	PS
FA18	MATH 241 CL2	4.0	A-
GS19	MATH 415 M16	3.0	A
GS20	MATH 231 AD2	3.0	A

3) PHYSICS SEQUENCE

SP20	PHYS 212 A11	4.0	A+
FA20	PHYS 211 A2	4.0	A+

4) SCIENCE ELECTIVE (COURSE LIST ON DEPARTMENT WEBSITE)

FA18	FSHN 120 A	3.0	A
------	------------	-----	---

5) COMPUTER SCIENCE REQUIREMENTS

NEEDS: 2 COURSES

SELECT FROM: CS 374, CS 421

Schedule

Schedule Details

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
00							
01							
02		CS374-ADC 02:55-03:15 (Pending)	CS421-D4 02:55-03:15 (Pending)				
03							
04							
05							

Panels

Summary

Title	Details	Hour	CRN	Schedule Typ	Status	Action
Intro to Algs & Mode...	CS 374, ADC	0	70645	Discussion...	Registered	Remove
Progrmg Languages...	CS 421, D4	4	40087	Lecture-Di...	Registered	Remove

Total Hours | Registered: 0 | Billing: 0 | CEU: 0 | Min: 0 | Max: 0

Submit



1) ORIENTATION AND PROFESSIONAL DEVELOPMENT

FA18	ENG 100 CS8	0.0	S
FA19	ECE 316 E2	3.0	A+



2) CORE MATHEMATICS COURSES

FA18	MATH 220 1	4.0	PS
FA18	MATH 241 CL2	4.0	A-
GS19	MATH 415 M16	3.0	A
GS20	MATH 231 AD2	3.0	A



3) PHYSICS SEQUENCE

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FA20	PHYS 211 A2	4.0	A+



4) SCIENCE ELECTIVE (COURSE LIST ON DEPARTMENT WEBSITE)

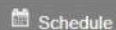
FA18	FSHN 120 A	3.0	A
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5) COMPUTER SCIENCE REQUIREMENTS

NEEDS: 2 COURSES

SELECT FROM: CS 374, CS 421



Schedule



Schedule Details

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
00							
01							
02		CS374-ADC 02:55-03:15 (Tuesday)	CS421-D4 02:55-03:15 (Tuesday)				
03							
04							
05							

Panels ▾

Summary

Title	Details	Hour	CRN	Schedule Typ	Status	Action	⚙
Intro to Algs & Mode...	CS 374, ADC	0	70645	Discussion...	Registered	Remove	▾
Progrmg Languages...	CS 421, D4	4	40087	Lecture-Di...	Registered	Remove	▾

Total Hours | Registered: 0 | Billing: 0 | CEU: 0 | Min: 0 | Max: 0

Submit

Grading

Warmups - 5%

Midterm Project - 40%

Final Project - 30%

Attendance - 15%

Quizzes - 10%

Grading

10% Class attendance

40% Homework

50% Project (35% presentation + 15% final report)

^	Other	Grade does not count toward the student's GPA or earned hours.
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ABS	Absent	More than six total unexcused absences or absent from the final exam without an acceptable excuse counts as a failure not acceptable for degree credit).
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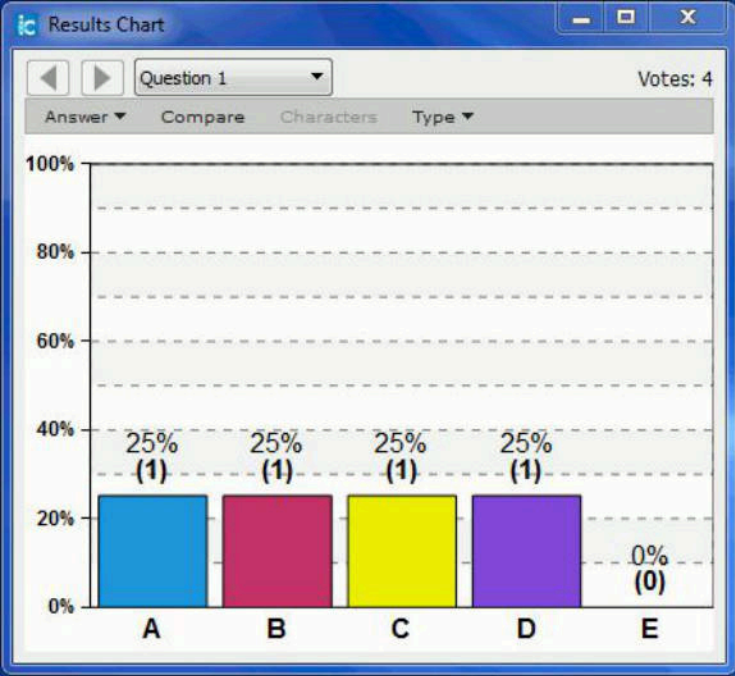
Δ II*	Audit	Indicates attendance as a visitor only.
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Other Grade does not count toward the student's GPA or earned hours.

ABS Absent More than six total unexcused absences or absent from the final exam without an acceptable excuse counts as a failure not acceptable for degree credit).

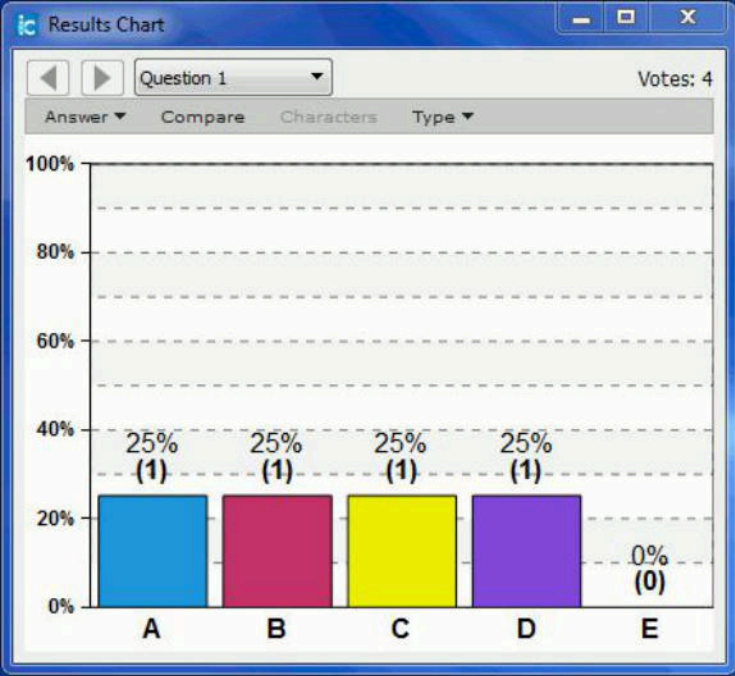
AII* Audit Indicates attendance as a visitor only

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	
00								
01								
02		CS374-ACC 02:00-03:15 (Pending)	CS421-04 02:15-03:30 (Pending)					
03								
04								
05								



DEMO 1





2020

JANUARY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

FEBRUARY

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

MARCH

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

APRIL

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

MAY

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

JUNE

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

JULY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

AUGUST

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

2020

JANUARY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

FEBRUARY

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

MARCH

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

APRIL

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

MAY

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

JUNE

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

JULY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

AUGUST

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

2020

JANUARY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

FEBRUARY

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

MARCH

S	M	T	W	T	F	S
1	2	3	?	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

APRIL

S	M	T	W	T	F	S
			?	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

MAY

S	M	T	W	T	F	S
			?		1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

JUNE

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

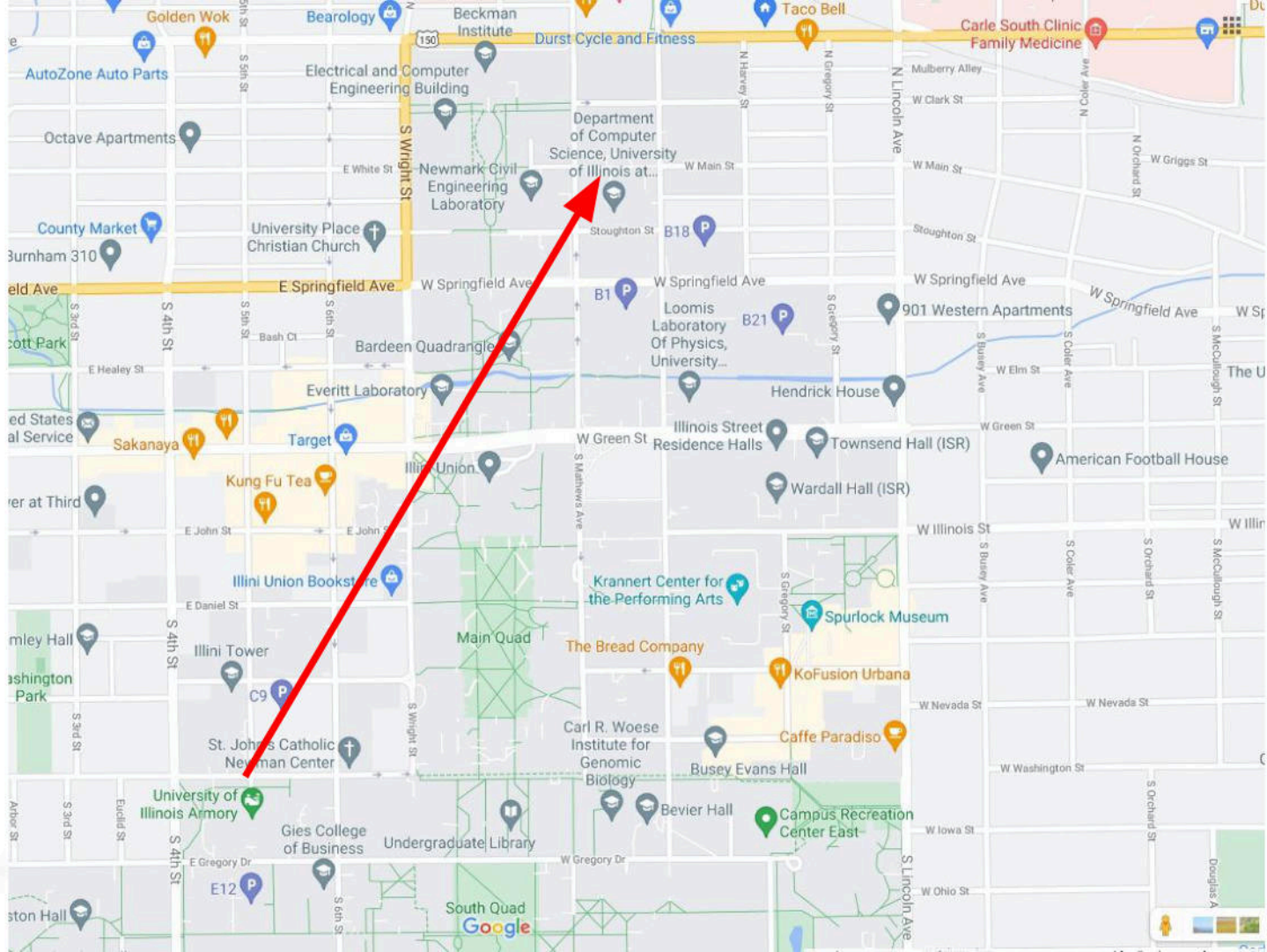
JULY

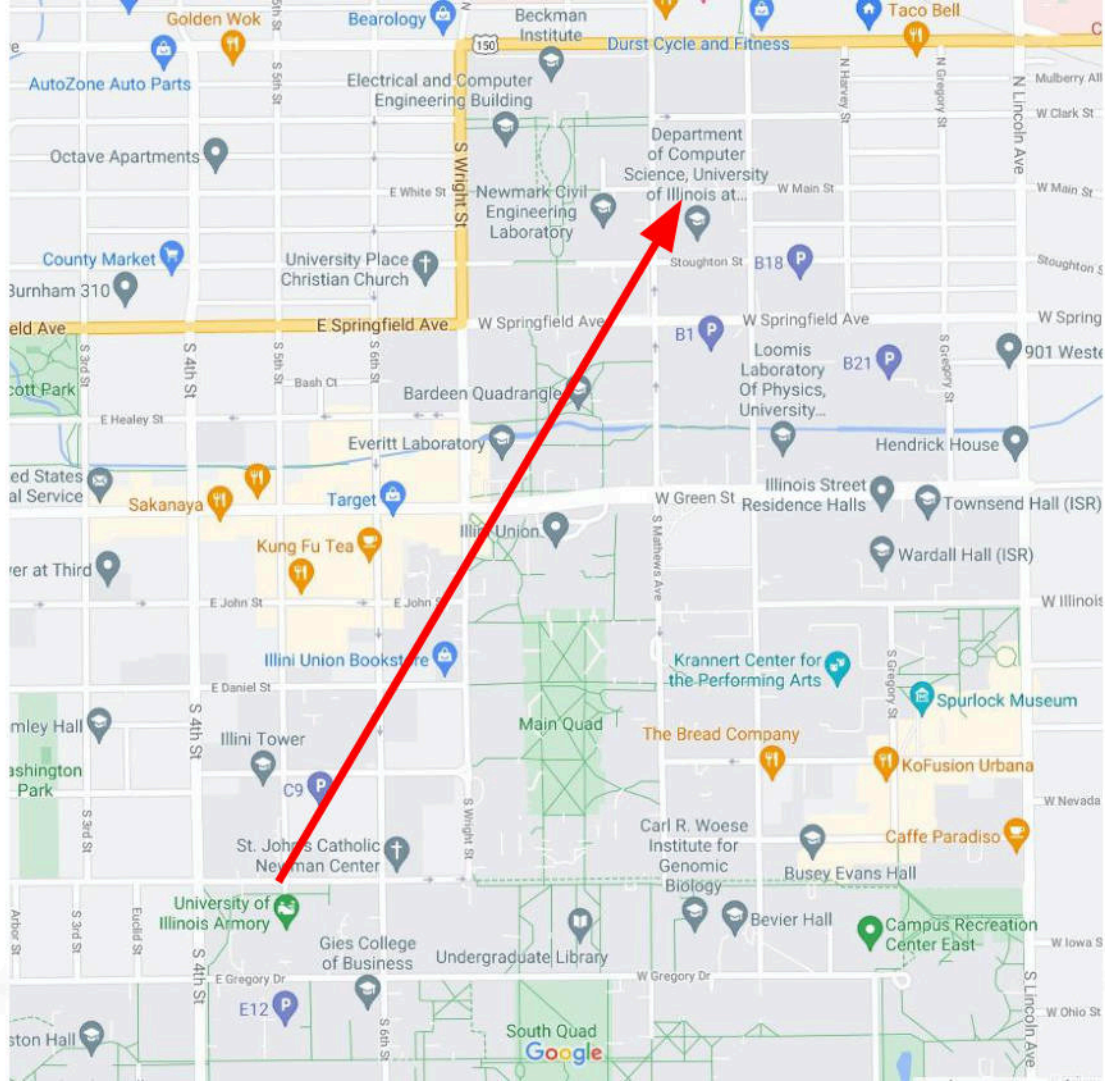
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26	27	28	29	30	31	

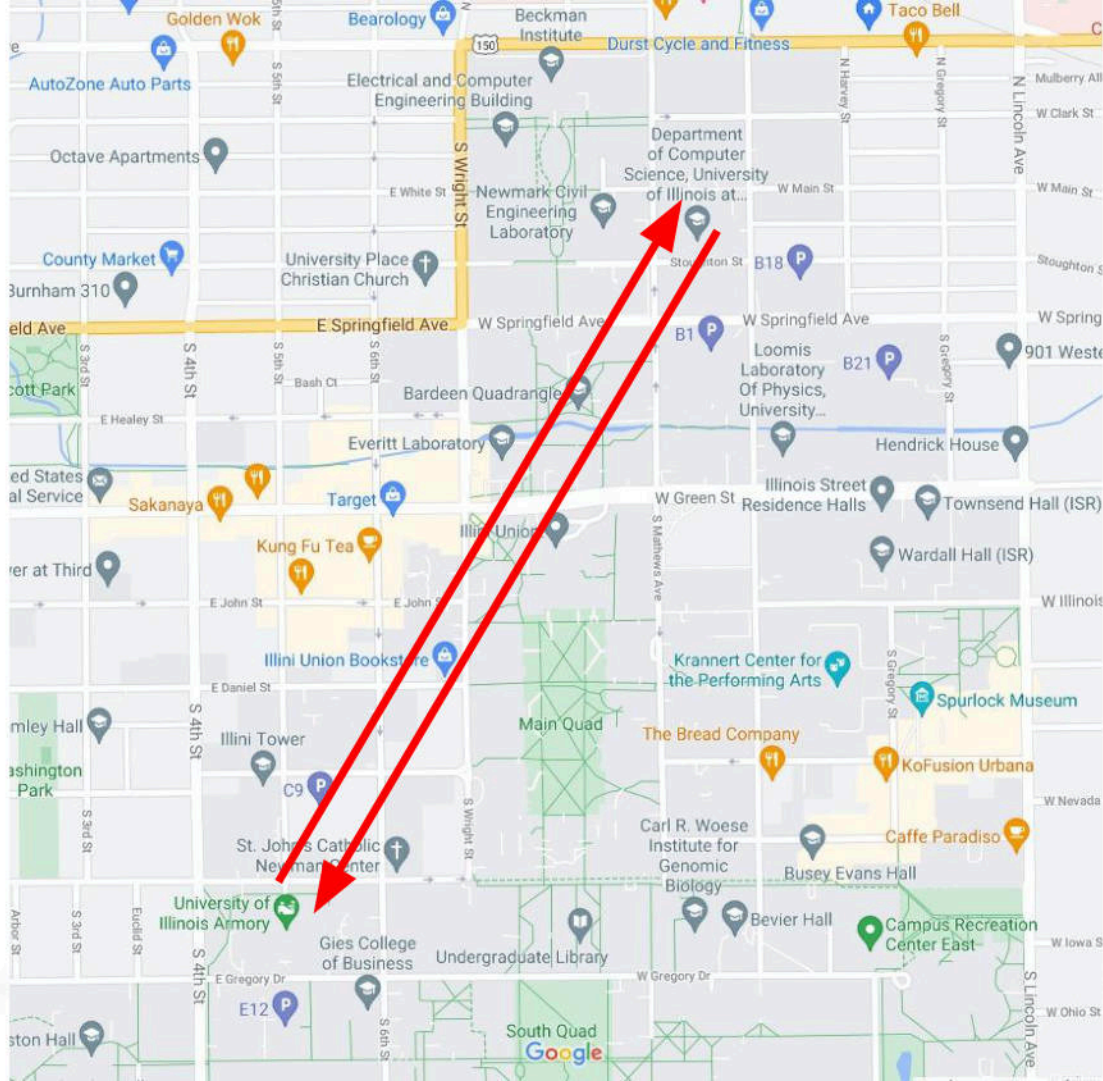
AUGUST

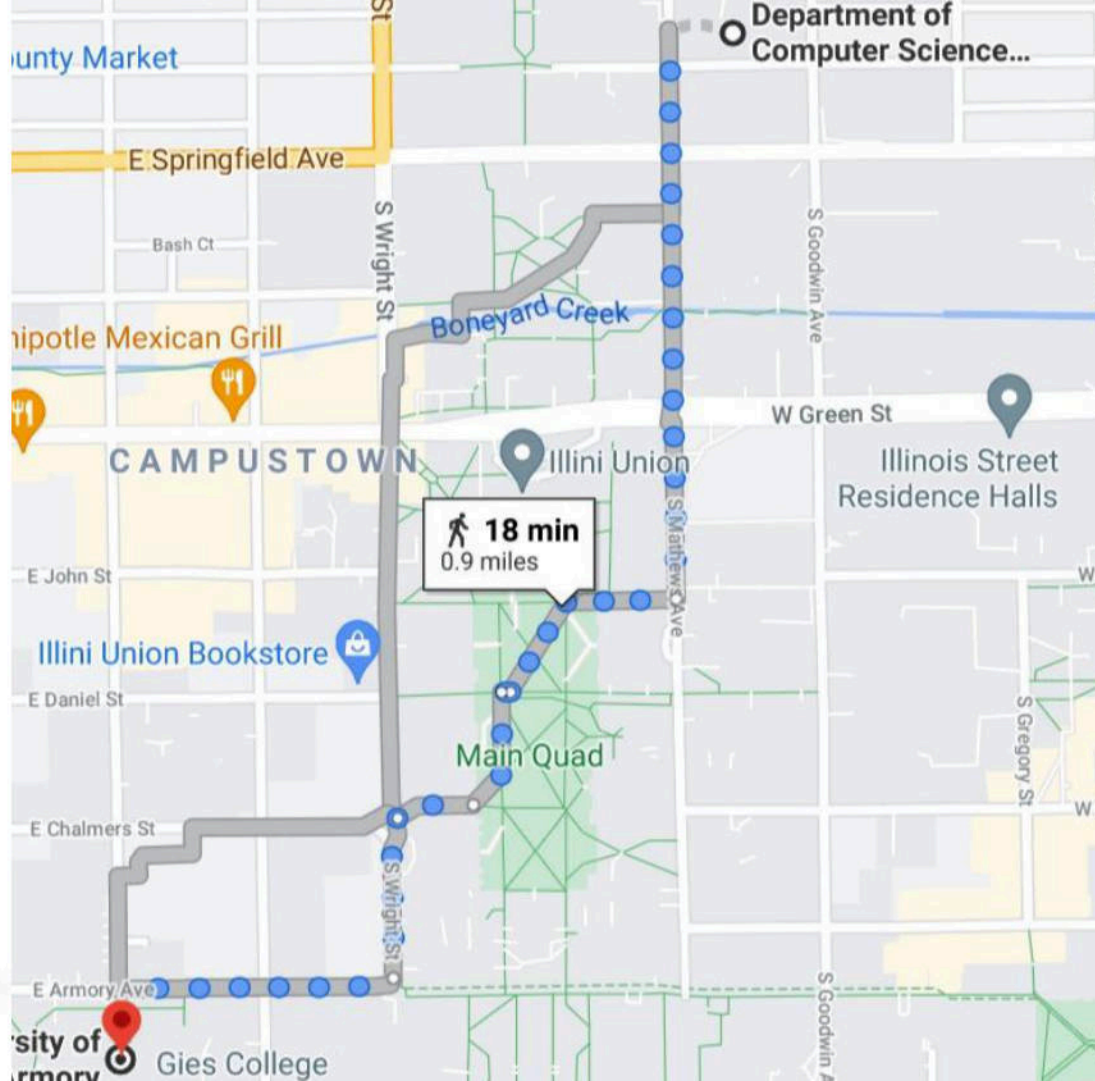
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2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					





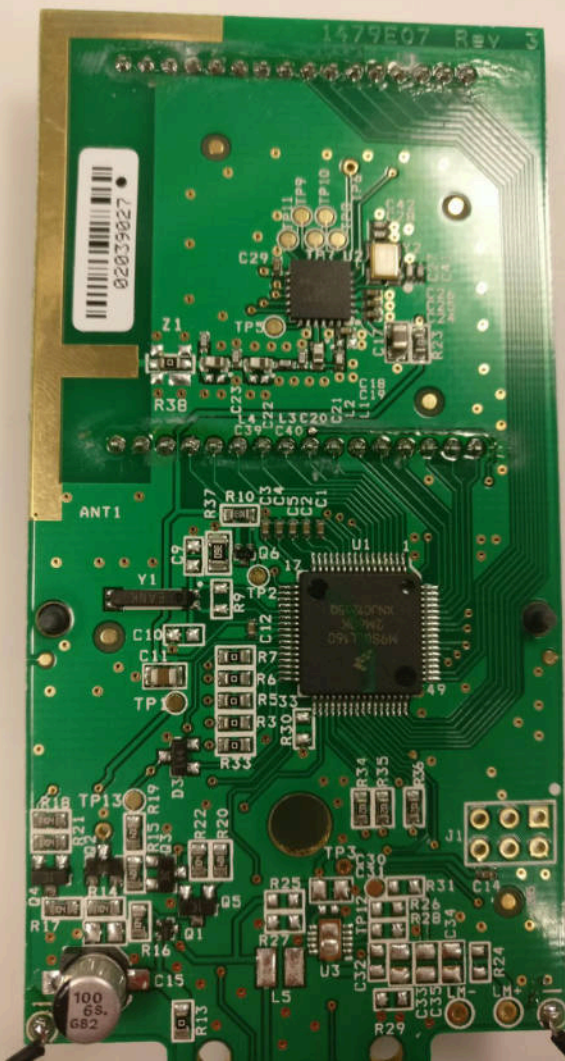
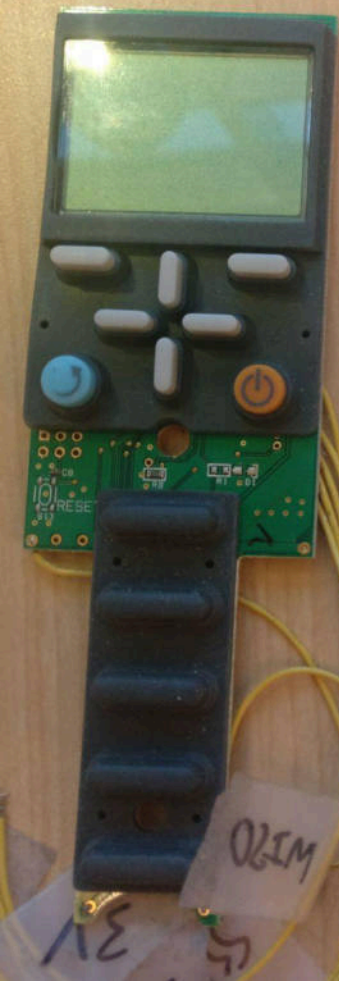










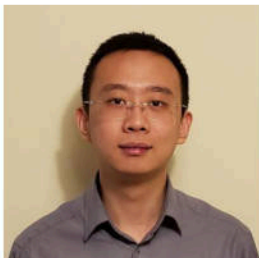




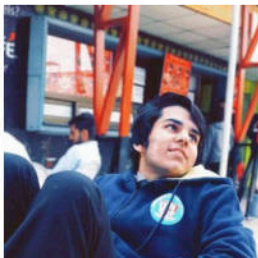
Aaron Wisner



Jacob Glueck



Charles Cao



Ammar Askar



Cole Smith



Prof. Bruce Land
(Advisor)

<https://github.com/wizard97/iSkipper>

<https://github.com/charlescao460/iSkipper-Software>

1

Reverse Engineering Classroom Polling: A Case Study

Daniel Wisner III, Jacob Glueck
developed 2/2017-5/2017, published 5/2019

Abstract—The [REDACTED] polling system is a widely used classroom voting system, which uses transmits the students' answers to the professor using an off the shelf 900 MHz radio. In this article, we completely reverse engineer the protocol used to transmit the answers, allowing an attacker to snoop on submitted answers, change students' answers by resubmitting them, and completely disable the system by flooding it rapidly with fake answers.

Index Terms—Classroom Voting, Reverse Engineering, Security

1 INTRODUCTION

IN this case study, we investigate the security of the most commonly deployed in-class polling system in the US, the [REDACTED] by attempting to reverse engineer it.

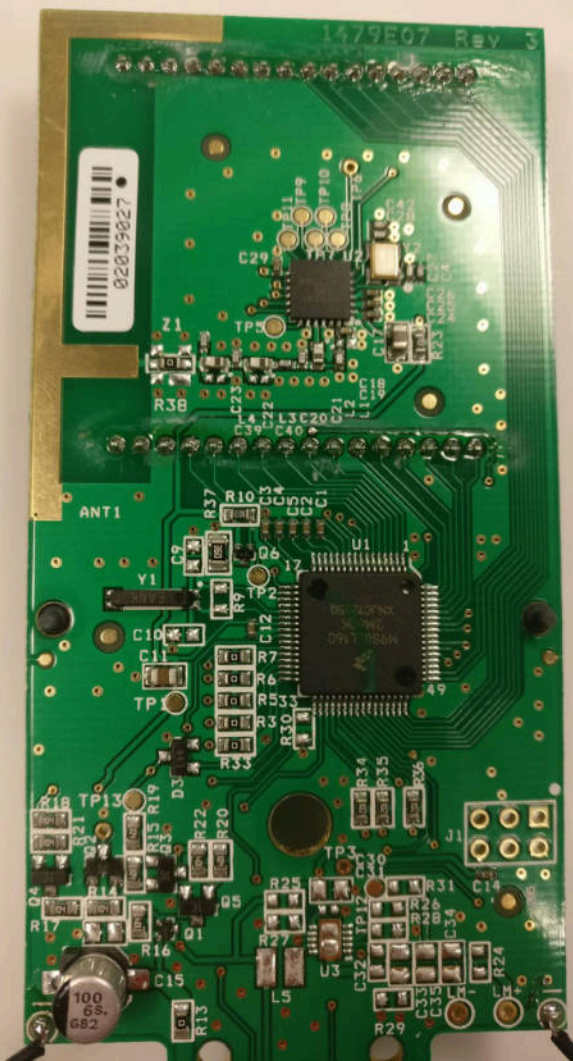
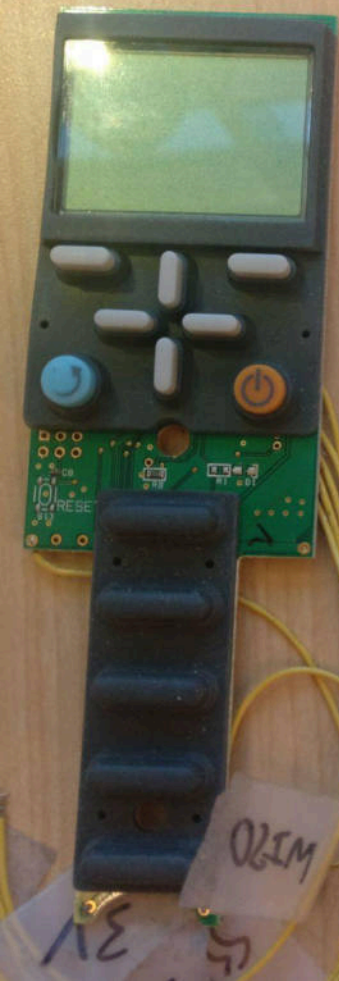
While a major portion of this paper explores the reverse engineering process of an off the shelf commercial device, this was done so to investigate its security and create/demonstrate practical exploits that we believe make the device unsuitable for use in its intended application. We propose several exploits a nefarious individual with the correct equipment can perform that take advantage of inherit vulnerabilities, these exploits include:

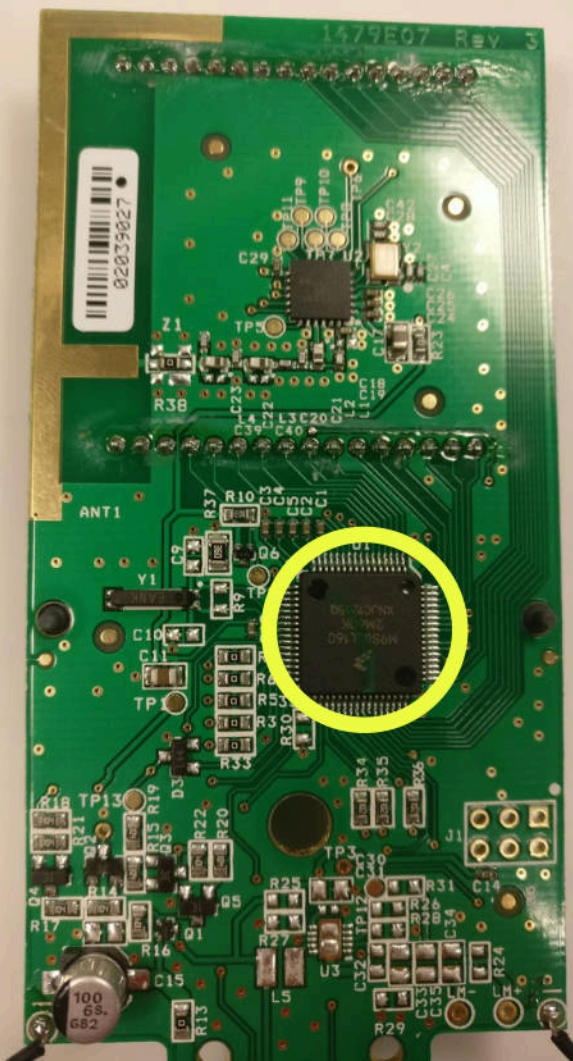
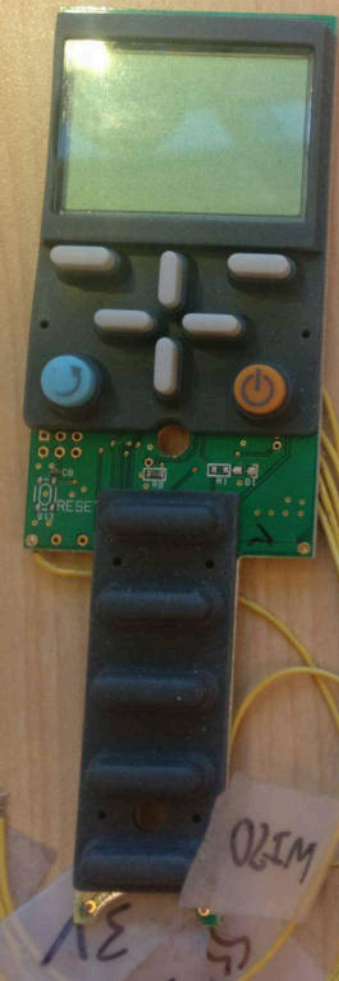
- Changing other respondees answers
- Creating an auto-answering device

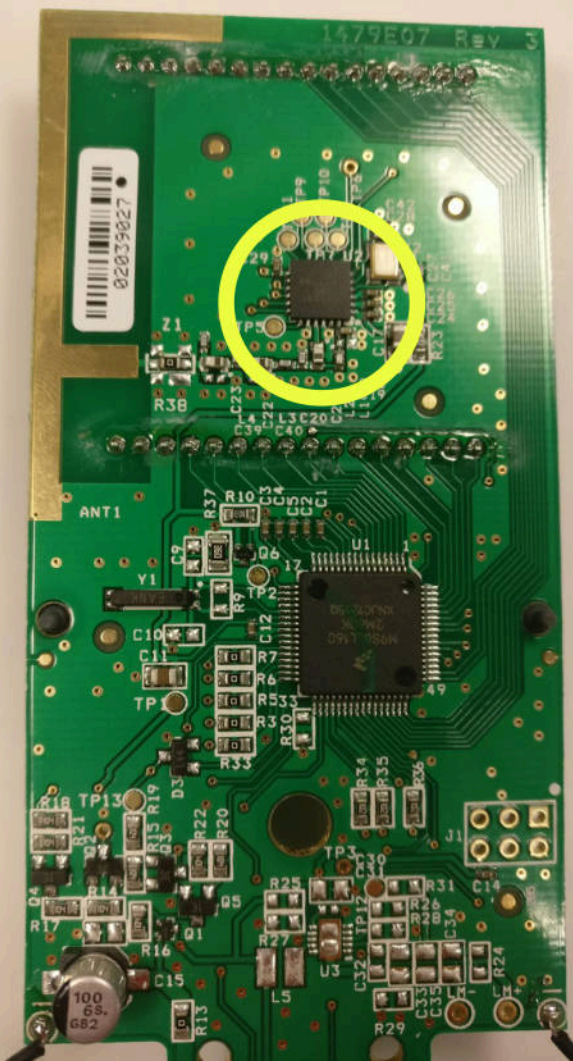
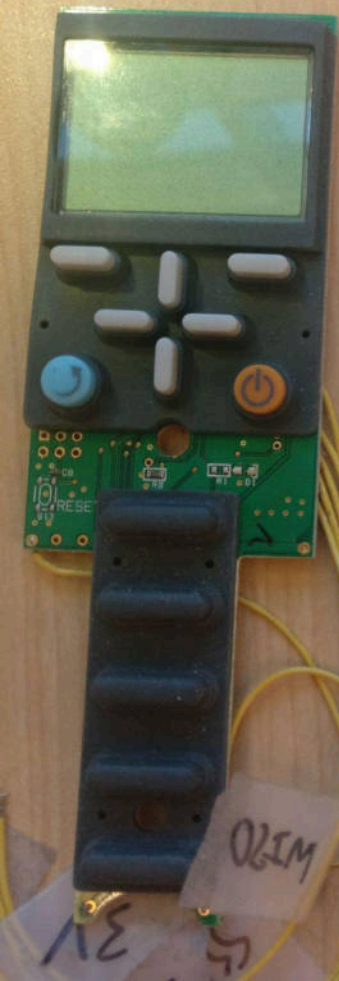
spent several several seconds submitting all 256 possible choices.

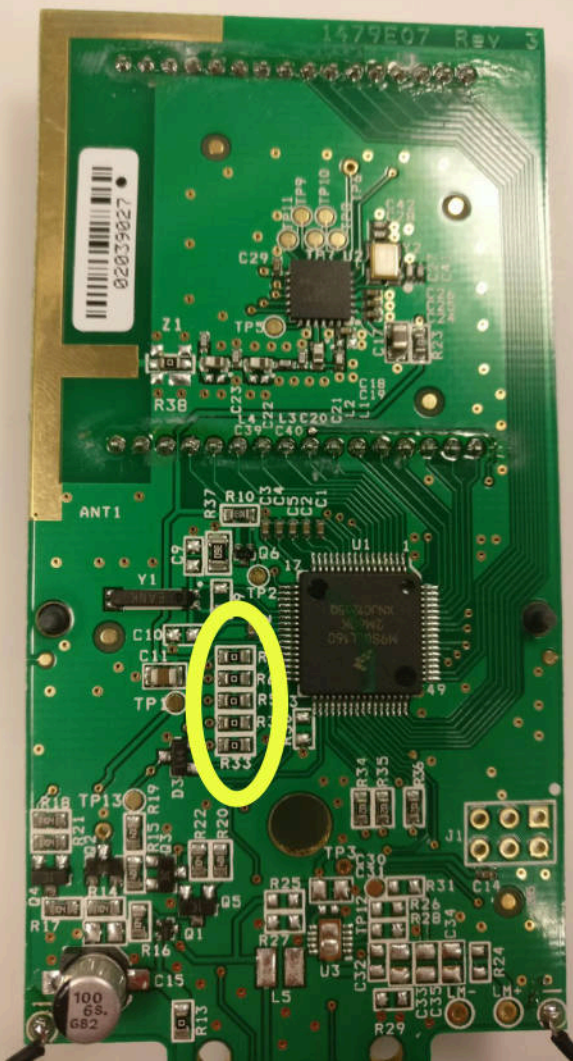
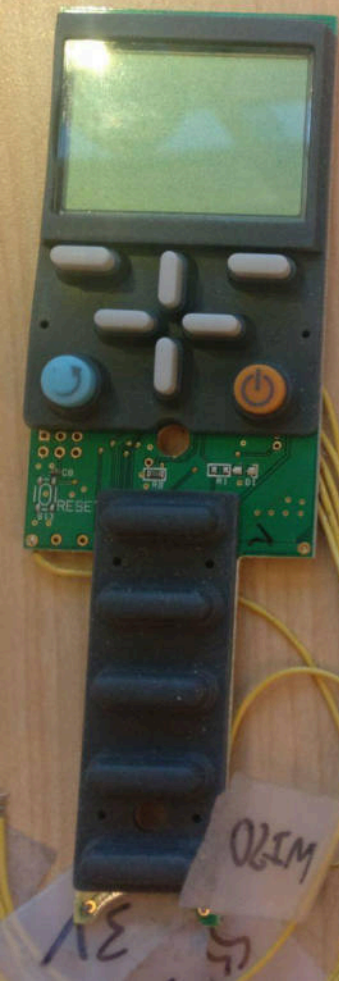
2 BACKGROUND

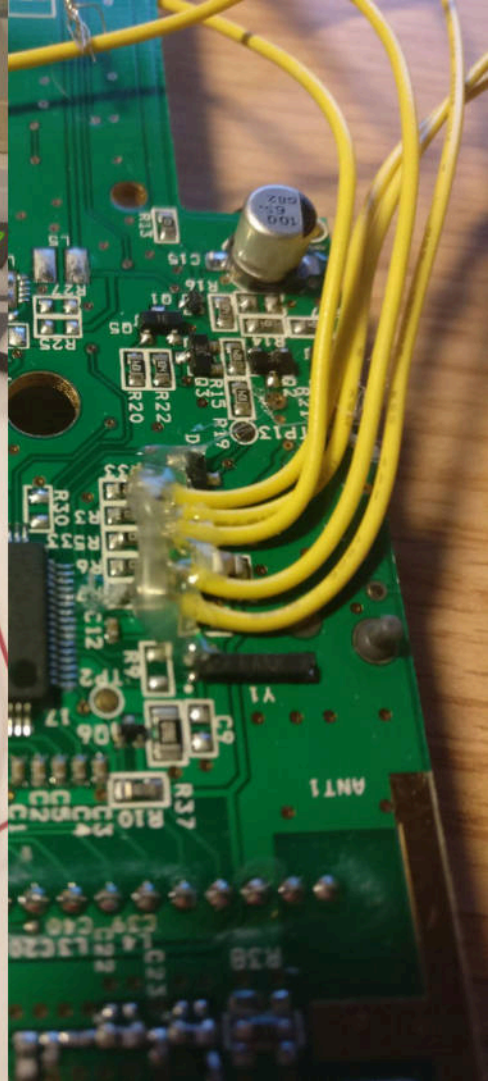
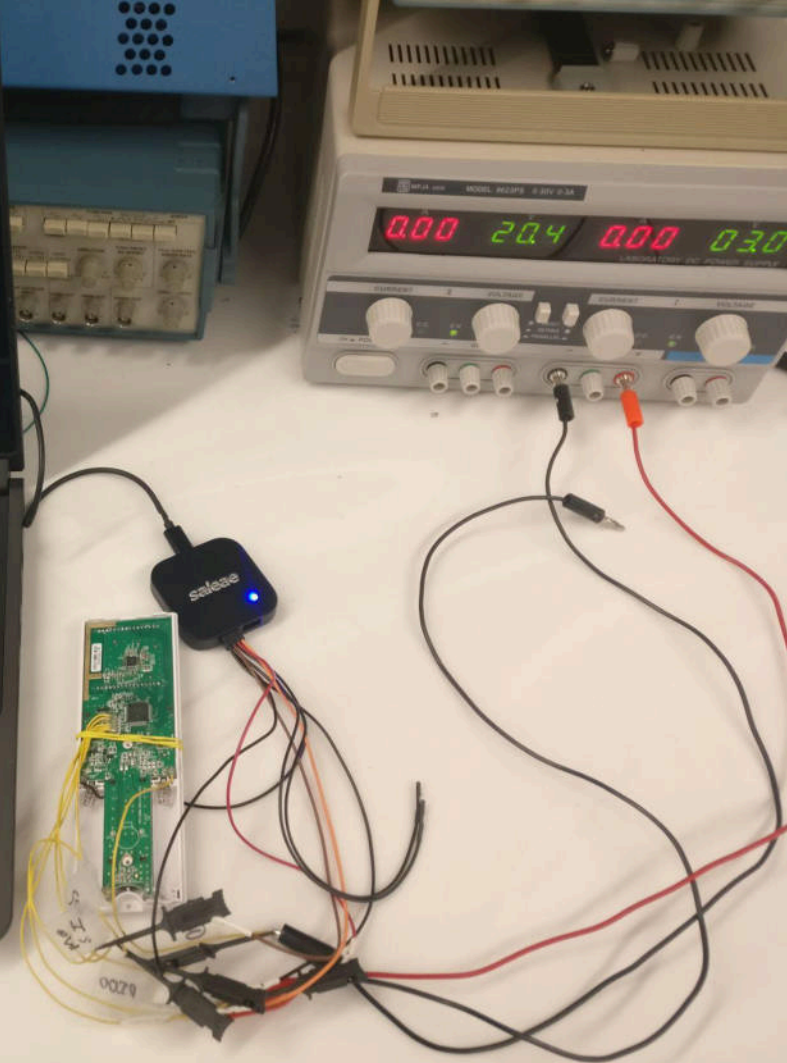
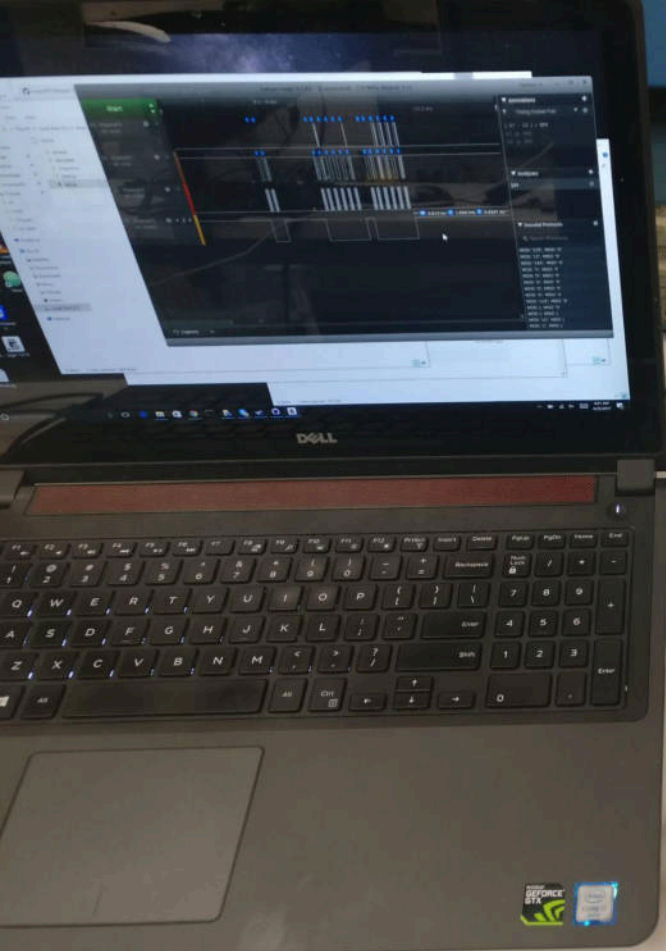
Universities across the globe have adopted electronic in-class polling devices for use in lecture-based classes. Generally, each student in the lecture possesses a wireless remote assigned to him/her, and throughout the lecture, a lecturer can pose questions to the audience that a student can respond to with his/her personal remote. A central base station, managed by the lecturer, receives these audiences' responses and records each student's response. [3]

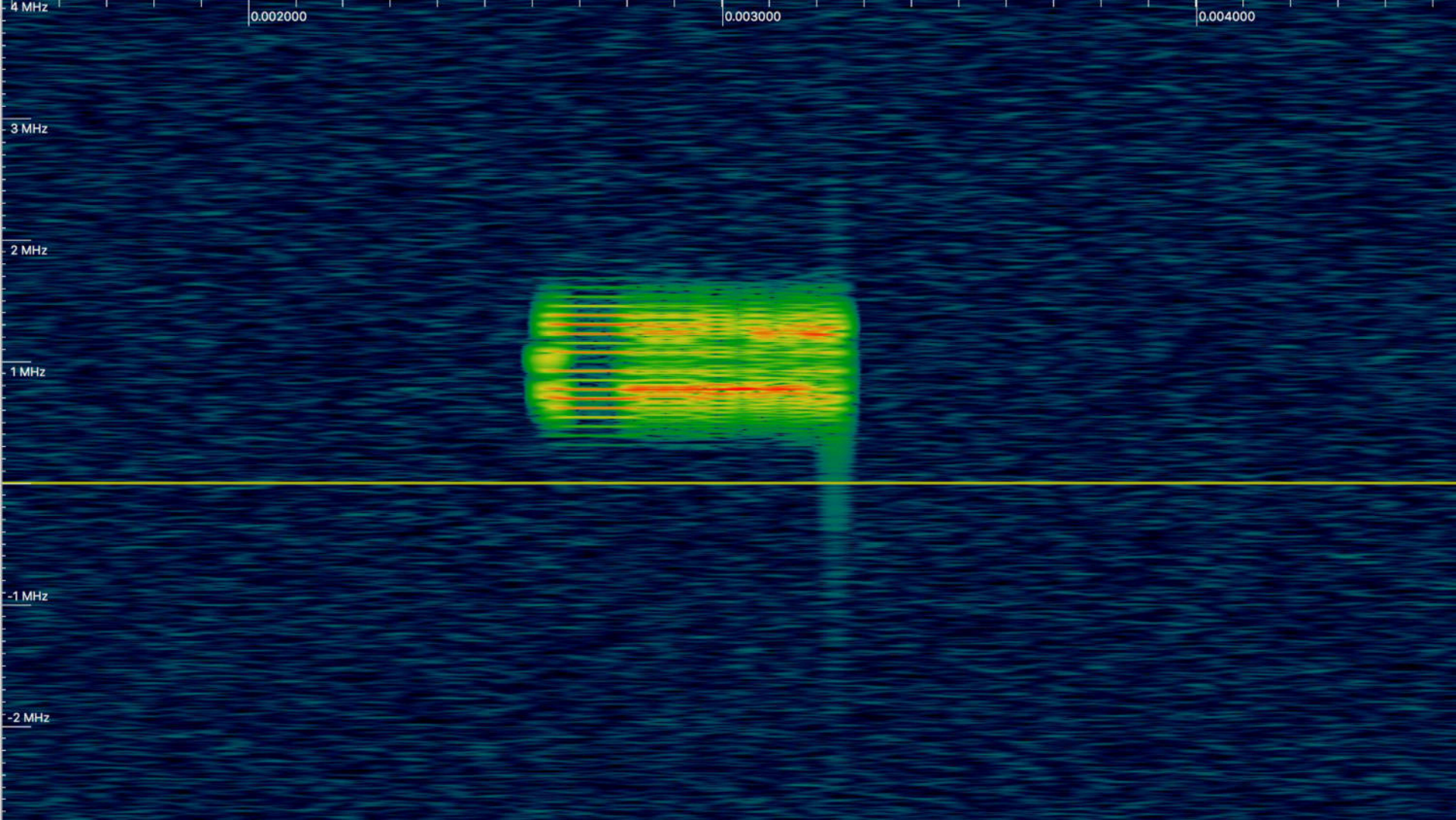


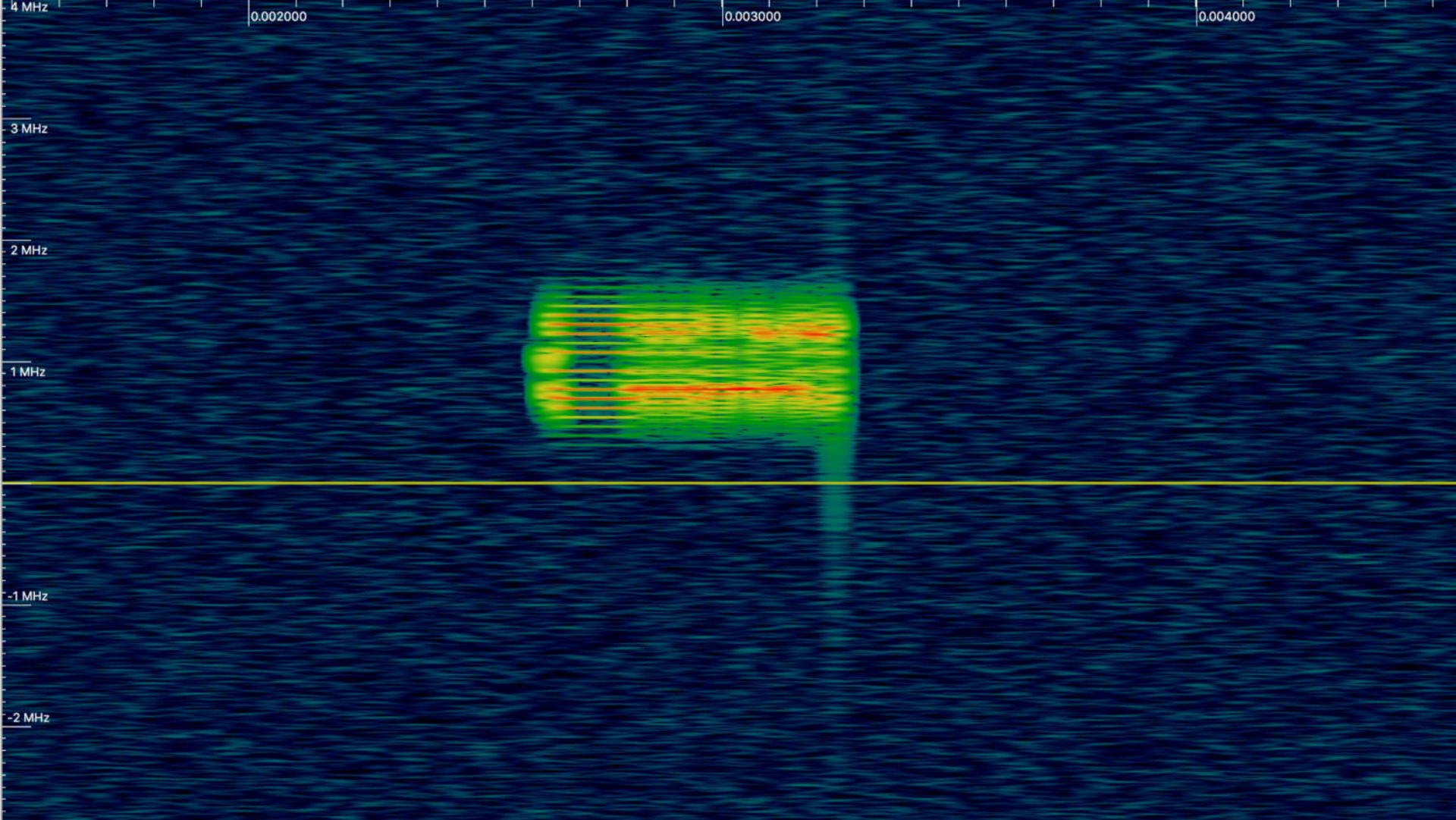












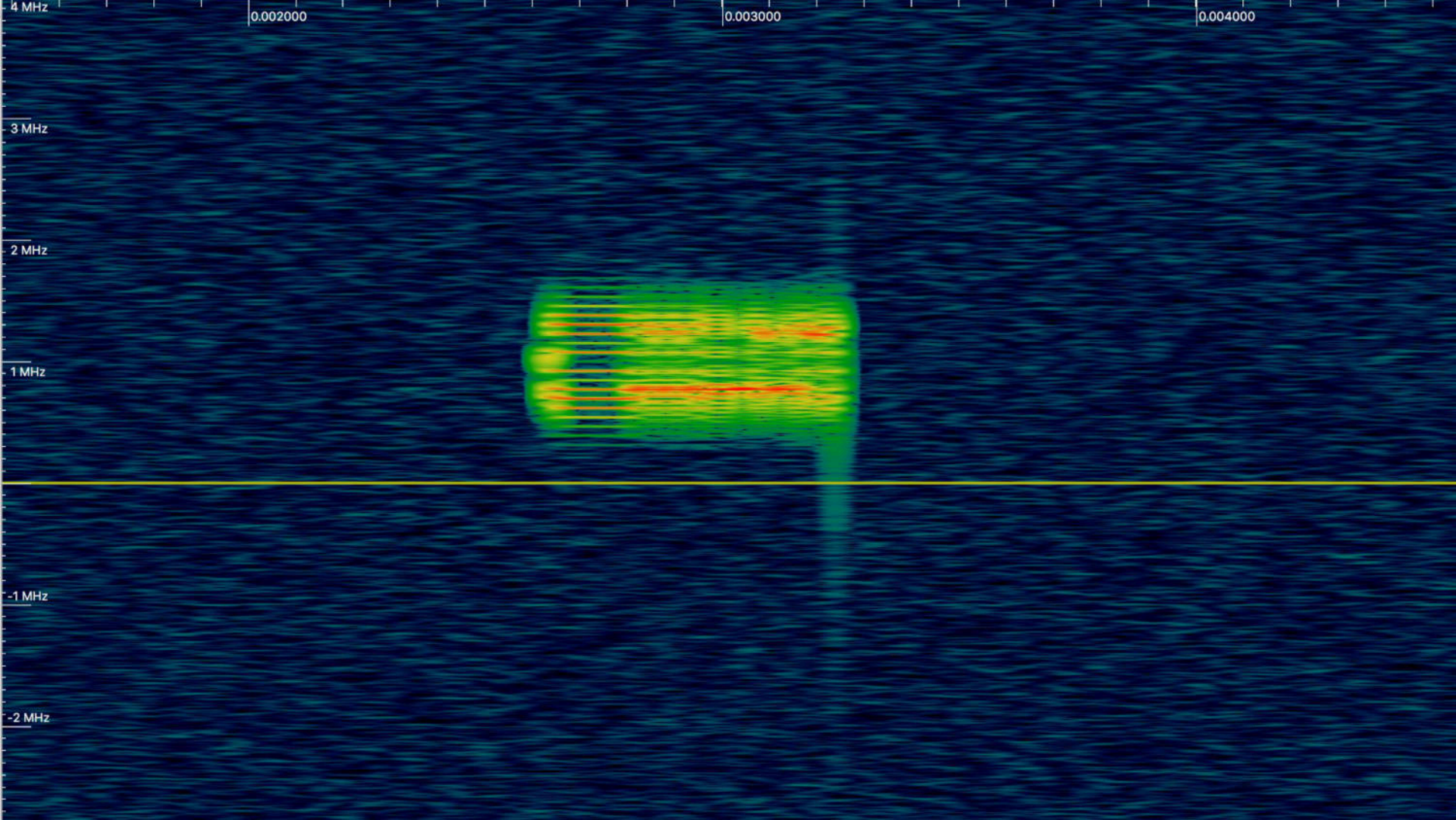


TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
AD	914.47	906.98
BA	915.47	907.98
BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
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AB	912.47	903.98
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BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 3
ID Encoding

(Transposition Cipher)

Byte 0	$I_0[4]$	$I_0[3]$	$I_0[2]$	$I_0[1]$	$I_0[0]$	$I_0[7]$	0	$I_1[7]$
Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

Device ID Checksum:

8F941803

$$0x8F \oplus 0x94 \oplus 0x18 = 0x03$$

TABLE 3
ID Encoding

(Transposition Cipher)

Byte 0	$I_0[4]$	$I_0[3]$	$I_0[2]$	$I_0[1]$	$I_0[0]$	$I_0[7]$	0	$I_1[7]$
Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
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C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

TABLE 4
4th Byte Least Significant Nibble Answer Encoding

Answer	Least Significant Nibble of 4th Byte
A	0x1
B	0x5
C	0xd
D	0xe
E	0xa

TABLE 1
Answer Packet Contents

Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

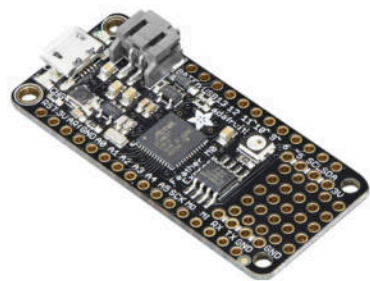
TABLE 6
Last Byte of Answer Packet

Answer	Last Byte of Answer Packet
A	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 1$
B	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 5$
C	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 13$
D	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 14$
E	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 10$
Ping	$\text{bytesum}(\text{packet}[0 : 5]) \bmod 256 + 2$

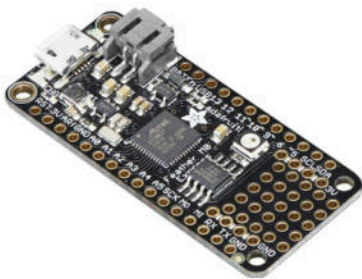
4 XXXXXXXXXX PROTOCOL

With the packet encoding figured out, the answer submission process can concisely be described as follows:

- 1) Generate first 3 bytes and 1 nibble of answer packet by encoding the 4 byte remote ID using Table 3
- 2) Set the least significant nibble of the 4th byte based on answer choice using Table 4
- 3) Set the 5th byte based on answer using Table 6
- 4) Transmit the following with 2-FSK modulation (frequency deviation of 222.833 KHz) at 152 kb/s at the transmit frequency defined in Table 2:
 - 3 bytes of preamble (0x55 or 0xAA)
 - The 3 byte sync address: 0x85, 0x85, 0x85
 - The encoded 5 byte answer packet created above
- 5) The base station will acknowledge receipt of the answer by sending back (ignoring the preamble) a 9 byte packet: the first 2 bytes are the first 2 bytes of the encoded remote ID (sync address), followed by an unknown 7-byte payload.



<https://github.com/VCNinc/Time-Turner>



```
#include <SPI.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include "IClickerEmulator.h"
#include <RingBufCPP.h>
#include <string.h>

#define MY_CLICKER_ID 0x0000AAAA
#define BUTTON_A 9
#define BUTTON_B 6
#define BUTTON_C 5
#define IS_RF69 true
#define IRQ_PIN 3
#define CSN 8
#define VBATPIN A7
#define MAX_BUFFERED_PACKETS 100
#define THRESHOLD 1000
#define MAX_RECVD 500
#define RAND_LOW 35
#define RAND_HIGH 75

uint8_t clicker_id[4];
int hij = 0;
int dos = 0;

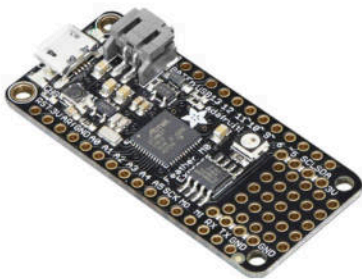
Adafruit_SSD1306 display = Adafruit_SSD1306(128, 32, &Wire);
IClickerAnswerPacket recvd[MAX_RECVD];
uint32_t num_recvd = 0;
IClickerEmulator clicker(CSN, IRQ_PIN, digitalPinToInterrupt(IRQ_PIN), IS_RF69);
RingBufCPP<IClickerPacket, MAX_BUFFERED_PACKETS> recvBuf;

int mode = 0;
int ans = 0;
bool active = false;
int ctr = 0;
bool a_state = true;
bool b_state = true;
bool c_state = true;
int sent = -1;
float measuredvbat = 0;
int batpercentage;

void input() {
  float measuredvbat = round((((float(analogRead(VBATPIN))/1024)*2*3.3) - 3.2) * 100);
  bool a_read = (measuredvbat < -140);
  bool b_read = digitalRead(BUTTON_B);
  bool c_read = digitalRead(BUTTON_C);
  if(a_read && a_read != a_state) {
    mode = (mode + 1) % 6;
    active = false;
  }
  if(b_read && b_read != b_state) {
    ans = (ans + 1) % 5;
  }
}
```

```
display.setTextColor(SSD1306_WHITE);
display.setCursor(100,24);
display.println(String(batpercentage) + "%");
display.setCursor(0,0);
switch(mode) {
  case 0:
    display.println("A Mode: View Votes");
    break;
  case 1:
    display.println("A Mode: Fake Votes");
    break;
  case 2:
    display.println("A Mode: Change Votes");
    break;
  case 3:
    display.println("A Mode: DoS Attack");
    break;
  case 4:
    display.println("A Mode: Copy Votes");
    break;
  case 5:
    display.println("A Mode: Vote Choice");
    break;
}
switch(ans) {
  case 0:
    display.println("B Choice: A");
    break;
  case 1:
    display.println("B Choice: B");
    break;
  case 2:
    display.println("B Choice: C");
    break;
  case 3:
    display.println("B Choice: D");
    break;
  case 4:
    display.println("B Choice: E");
    break;
}
if(active) {
  display.println("C Active: Yes");
  ctr = (++ctr) % 50;
  if (ctr == 0) sent = -1;
  if(mode == 0 || mode == 4 || mode == 5) {
    char tmp[100];
    uint16_t res[NUM_ANSWER_CHOICES] = { 0 };
    for (uint32_t i = 0; i < num_recvd; i++) {
      res[recvd[i].answer]++;
    }
    snprintf(tmp, sizeof(tmp), "A%u B%u C%u D%u E%u",
      if(mode == 5) {
        switch(ans) {
          case 0:
```

<https://github.com/VCNinc/Time-Turner>



```
#include <SPI.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include "IClickerEmulator.h"
#include <RingBufCPP.h>
#include <string.h>

#define MY_CLICKER_ID 0x0000AAAA
#define BUTTON_A 9
#define BUTTON_B 6
#define BUTTON_C 5
#define IS_RF69HW true
#define IRQ_PIN 3
#define CSN 8
#define VBATPIN A7
#define MAX_BUFFERED_PACKETS 100
#define THRESHOLD 1000
#define MAX_RECVD 500
#define RAND_LOW 35
#define RAND_HIGH 75

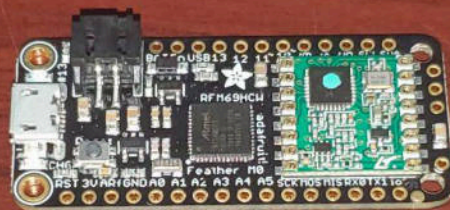
uint8_t clicker_id[4];
int hij = 0;
int dos = 0;

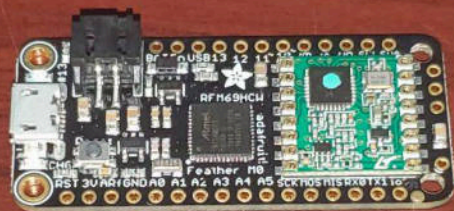
Adafruit_SSD1306 display = Adafruit_SSD1306(128, 32, &Wire);
IClickerAnswerPacket recvd[MAX_RECVD];
uint32_t num_recvd = 0;
IClickerEmulator clicker(CSN, IRQ_PIN, digitalPinToInterrupt(IRQ_PIN), IS_RF69HW);
RingBufCPP<IClickerPacket, MAX_BUFFERED_PACKETS> recvBuf;

int mode = 0;
int ans = 0;
bool active = false;
int ctr = 0;
bool a_state = true;
bool b_state = true;
bool c_state = true;
int sent = -1;
float measuredvbat = 0;
int batpercentage;

void input() {
  float measuredvbat = round((((float(analogRead(VBATPIN))/1024)*2*3.3) - 3.2) * 100);
  bool a_read = (measuredvbat < -140);
  bool b_read = digitalRead(BUTTON_B);
  bool c_read = digitalRead(BUTTON_C);
  if(a_read && a_read != a_state) {
    mode = (mode + 1) % 6;
    active = false;
  }
  if(b_read && b_read != b_state) {
    ans = (ans + 1) % 5;
  }
```

```
display.setTextColor(SSD1306_WHITE);
display.setCursor(100,24);
display.println(String(batpercentage) + "%");
display.setCursor(0,0);
switch(mode) {
  case 0:
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    break;
  case 1:
    display.println("A Mode: Fake Votes");
    break;
  case 2:
    display.println("A Mode: Change Votes");
    break;
  case 3:
    display.println("A Mode: DoS Attack");
    break;
  case 4:
    display.println("A Mode: Copy Votes");
    break;
  case 5:
    display.println("A Mode: Vote Choice");
    break;
}
switch(ans) {
  case 0:
    display.println("B Choice: A");
    break;
  case 1:
    display.println("B Choice: B");
    break;
  case 2:
    display.println("B Choice: C");
    break;
  case 3:
    display.println("B Choice: D");
    break;
  case 4:
    display.println("B Choice: E");
    break;
}
if(active) {
  display.println("C Active: Yes");
  ctr = (++ctr) % 50;
  if (ctr == 0) sent = -1;
  if(mode == 0 || mode == 4 || mode == 5) {
    char tmp[100];
    uint16_t res[NUM_ANSWER_CHOICES] = { 0 };
    for (uint32_t i = 0; i < num_recvd; i++) {
      res[recvd[i].answer]++;
    }
    snprintf(tmp, sizeof(tmp), "A%u B%u C%u D%u E%u",
      if(mode == 5) {
        switch(ans) {
          case 0:
```







i> Skipper

Multiple

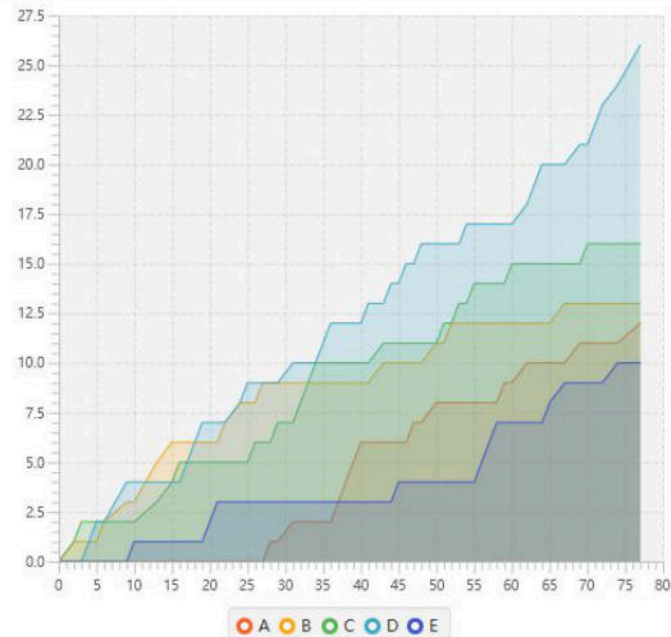
DEMO 3

Status: **Capturing** Channel: AA
ID: CDCDCDCD

ID Count: 77

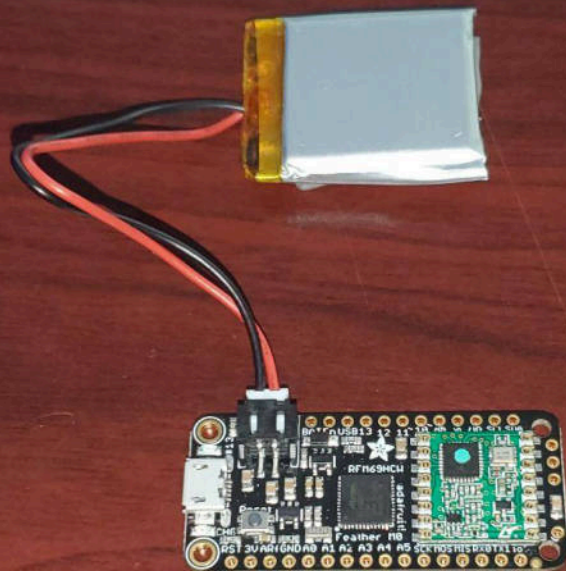
Response Count: 77

A:12 B:13 C:16 D:26 E:10



“Time Turner” Protocol

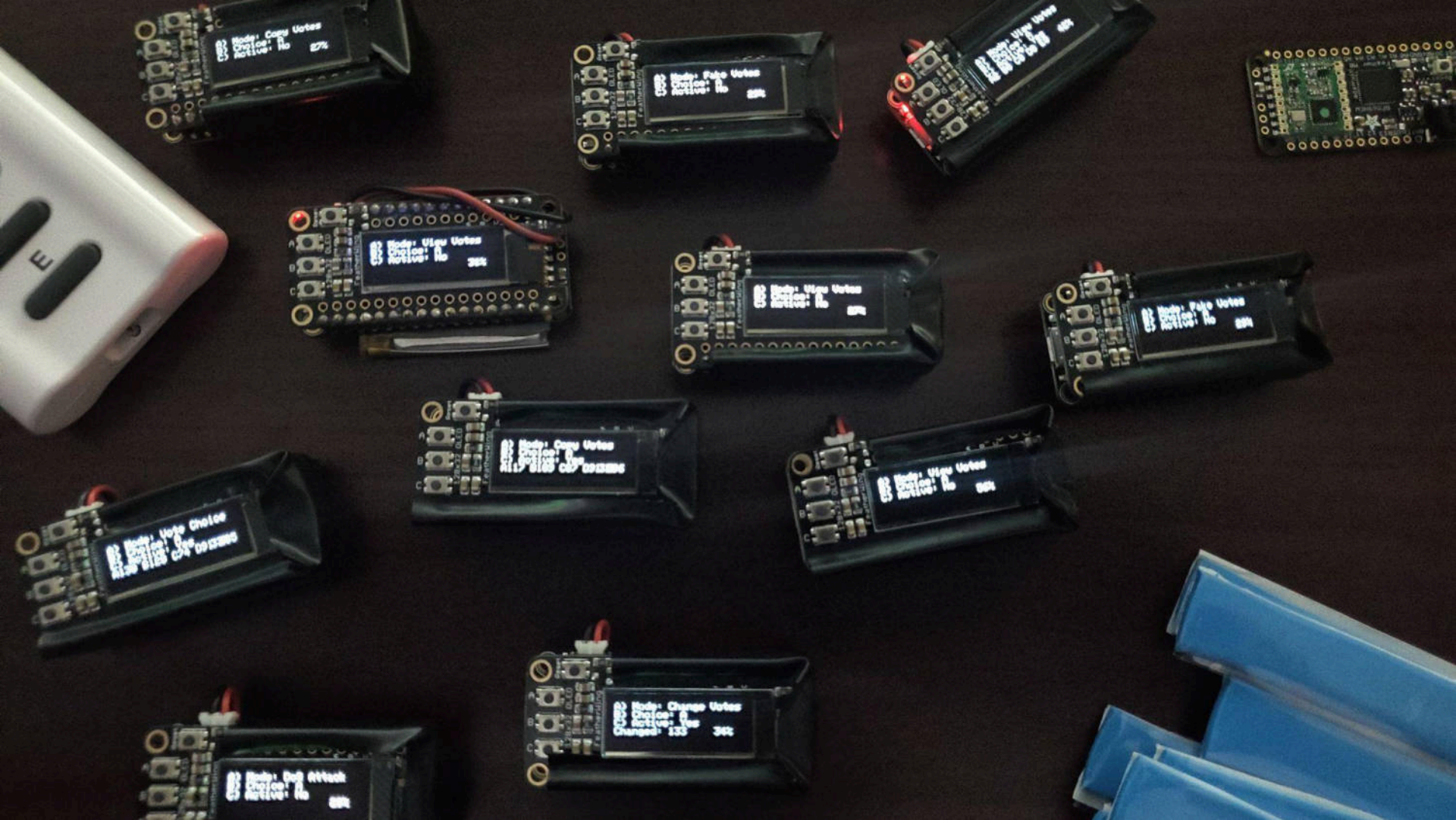
1. Remain idle until the class is about to begin
2. Turn on and start emulating both a remote and a base station
3. Wait until device overhears a burst of radio traffic
4. Determine most popular student response using base station emulator
5. Broadcast most popular response from remote emulator
6. Repeat 2-5 until the expected end time of the class











02 Mode: CPU Util
03 Action: No
27%

02 Mode: CPU Util
03 Action: No
28%

02 Mode: CPU Util
03 Action: No
29%

02 Mode: CPU Util
03 Action: No
30%

02 Mode: CPU Util
03 Action: No
31%

02 Mode: CPU Util
03 Action: No
32%

02 Mode: CPU Util
03 Action: No
33%

02 Mode: CPU Util
03 Action: No
34%

02 Mode: CPU Util
03 Action: No
35%

02 Mode: CPU Util
03 Action: No
36%

02 Mode: CPU Util
03 Action: No
37%

02 Mode: CPU Util
03 Action: No
38%

Assignment Name	Point Breakdown							
Bonus 95%	1. 100.0%	2. 100.0%	3. 0.0%	4. 0.0%	5. 100.0%	6. 100.0%	7. 100.0%	8. 100.0%
Checkpoints 100%	1. 100.0%	Bonus 95%				6. 100.0%	7. 100.0%	8. 100.0%
Discussions 95%	Quiz 1 0.0% (EX)					Quiz 6 100.0%	Quiz 7 100.0%	Quiz 8 100.0%
Exams 94%	1. 97.1%							
Exams - Scaled 94%	1. 97.1%	2. 87.0%	3. 93.6%	4. 96.8%				
Homework 98%	1. 100.0%	Attendance 96%				6. 97.5%	7. 99.5%	8. 98.7%
James Scholar	1.							
Labs 98%	1. 0.0% (EX)					6. 100.0%	7. 100.0%	8. 100.0%
Attendance 96%	1. 100.0%	2. 100.0%	3. 0.0%	4. 0.0%	5. 100.0%	6. 100.0%	7. 100.0%	8. 100.0%
Plectures 100%	1. 100.0%	2. 100.0%	3. 100.0%	4. 100.0%	5. 100.0%	6. 100.0%	7. 100.0%	8. 100.0%

Authentication

Authentication

An authentication mechanism can be:

Authentication

An authentication mechanism can be:

1. Something you know (eg. passwords)

Authentication

An authentication mechanism can be:

1. Something you know (eg. passwords)
2. Something you have (eg. U2F keys)

Authentication

An authentication mechanism can be:

1. Something you know (eg. passwords)
2. Something you have (eg. U2F keys)
3. Something you are (eg. biometrics)

Authentication

An authentication mechanism can be:

1. Something you know (eg. passwords)
2. Something you have (eg. U2F keys)
3. Something you are (eg. biometrics)
4. Polling devices???

TABLE 3
ID Encoding

Byte 0	$I_0[4]$	$I_0[3]$	$I_0[2]$	$I_0[1]$	$I_0[0]$	$I_0[7]$	0	$I_1[7]$
Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

Kerckhoffs's Principle

A system should be secure even if everything except the key is public knowledge.

TABLE 3
ID Encoding

Byte 0	$I_0[4]$	$I_0[3]$	$I_0[2]$	$I_0[1]$	$I_0[0]$	$I_0[7]$	0	$I_1[7]$
Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

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TABLE 3
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Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

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TABLE 3
ID Encoding

Byte 0	$I_0[4]$	$I_0[3]$	$I_0[2]$	$I_0[1]$	$I_0[0]$	$I_0[7]$	0	$I_1[7]$
Byte 1	$I_1[6]$	$I_1[5]$	$I_1[4]$	$I_1[3]$	$I_1[2]$	$I_1[1]$	0	$I_0[6]$
Byte 2	$I_1[0]$	$I_2[7]$	$I_2[6]$	$I_2[5]$	$I_2[4]$	$I_2[3]$	0	$I_0[5]$
Byte 3	$I_2[2]$	$I_2[1]$	$I_2[0]$	$I_2[0]$	0	0	0	0

CIA Properties

CIA Properties

- Confidentiality

CIA Properties

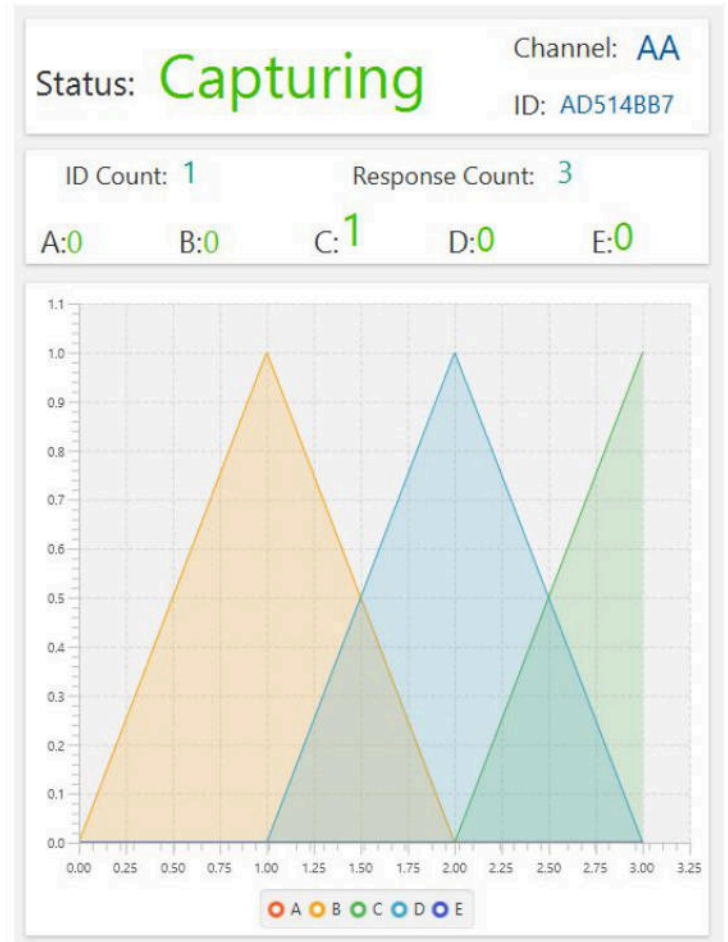
- Confidentiality
- Integrity

CIA Properties

- Confidentiality
- Integrity
- Availability

CIA Properties

- Confidentiality
- Integrity
- Availability



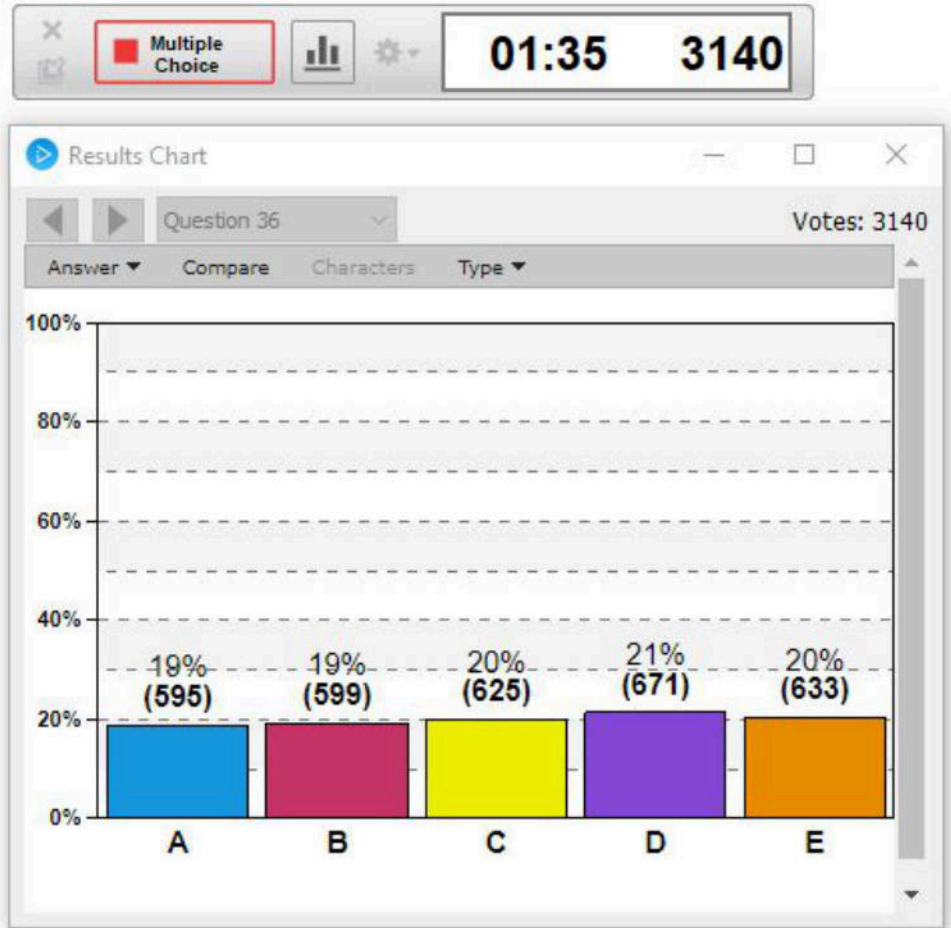
CIA Properties

- Confidentiality
- **Integrity**
- Availability



CIA Properties

- Confidentiality
- Integrity
- **Availability**



Availability

TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
AD	914.47	906.98
BA	915.47	907.98
BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

Availability

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Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
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BA	915.47	907.98
BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
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Availability

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Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
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AD	914.47	906.98
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BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

TABLE 1
Answer Packet Contents

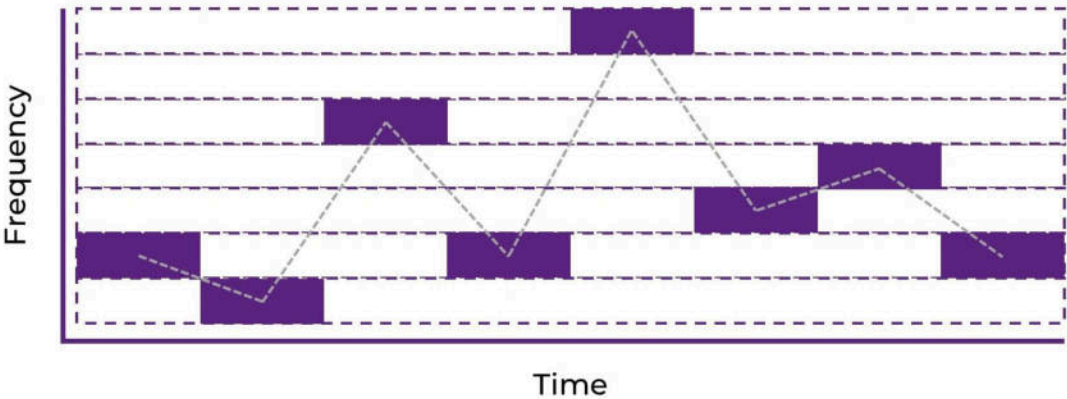
Answer	Packet Contents
A	0x7d, 0x28, 0x0c, 0x01, 0xb2
B	0x7d, 0x28, 0x0c, 0x05, 0xb6
C	0x7d, 0x28, 0x0c, 0x0d, 0xbe
D	0x7d, 0x28, 0x0c, 0x0e, 0xbf
E	0x7d, 0x28, 0x0c, 0x0a, 0xbb

Availability

TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
AD	914.47	906.98
BA	915.47	907.98
BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

Frequency-hopping spread spectrum

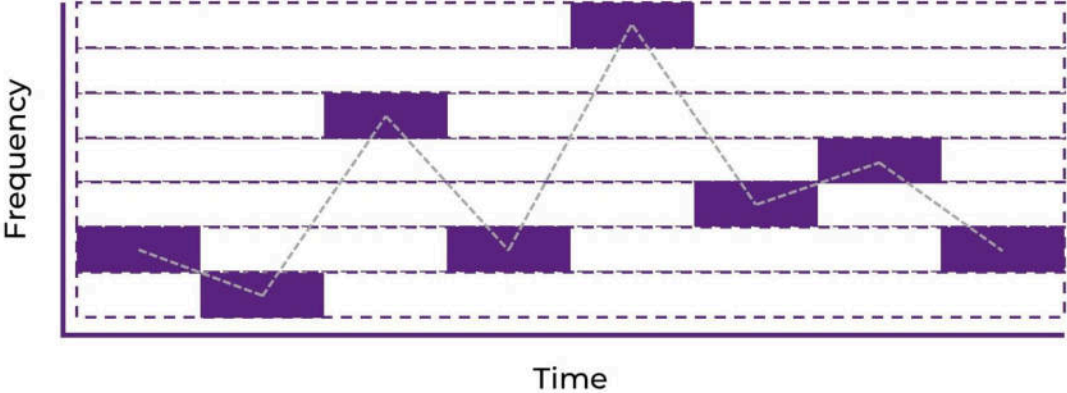


Availability

TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
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BC	919.47	910.97
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CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

Frequency-hopping spread spectrum

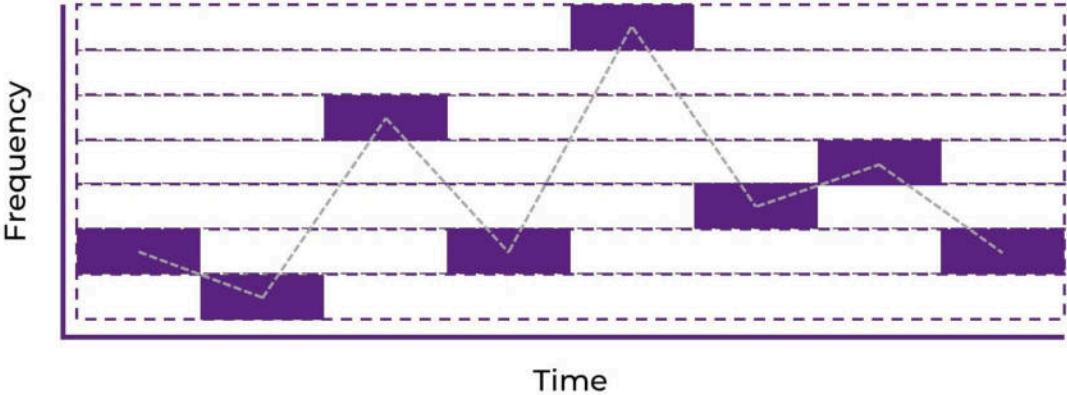


Availability

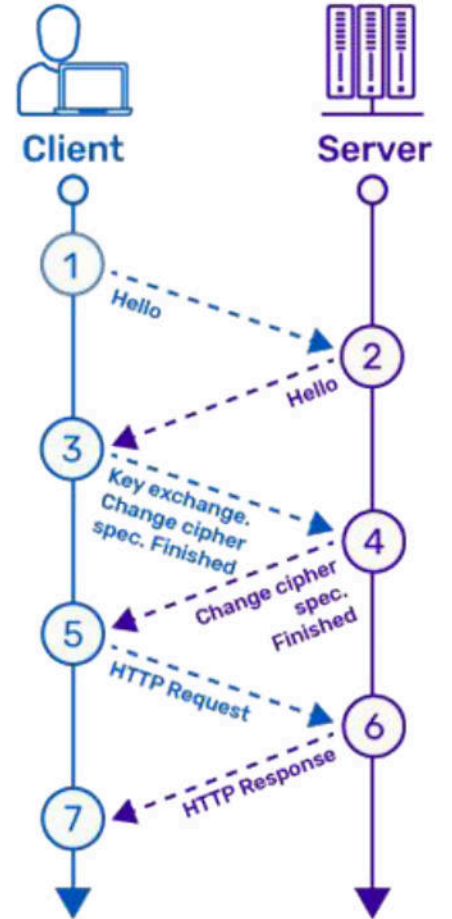
TABLE 2
Channel Frequencies

Channel	Transmit Frequency (MHz)	Receive Frequency (MHz)
AA	916.47	902.98
AB	912.47	903.98
AC	913.47	905.48
AD	914.47	906.98
BA	915.47	907.98
BB	918.48	909.48
BC	919.47	910.97
BD	920.47	911.97
CA	921.47	913.47
CB	922.47	914.97
CC	906.48	915.97
CD	907.48	917.47
DA	904.98	918.47
DB	908.48	919.97
DC	910.48	921.47
DD	909.48	922.47

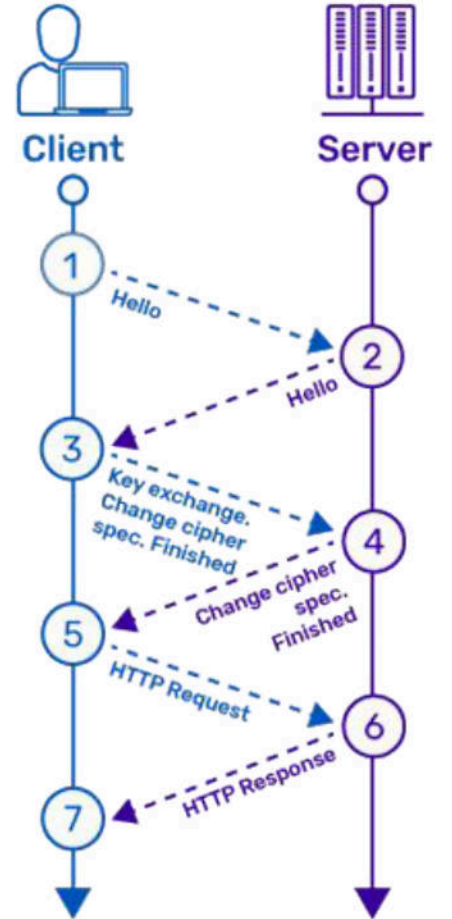
Frequency-hopping spread spectrum



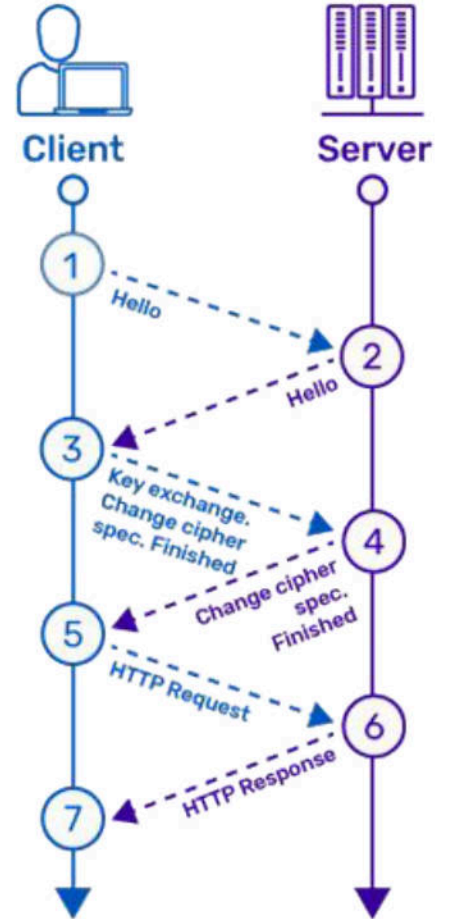
Confidentiality



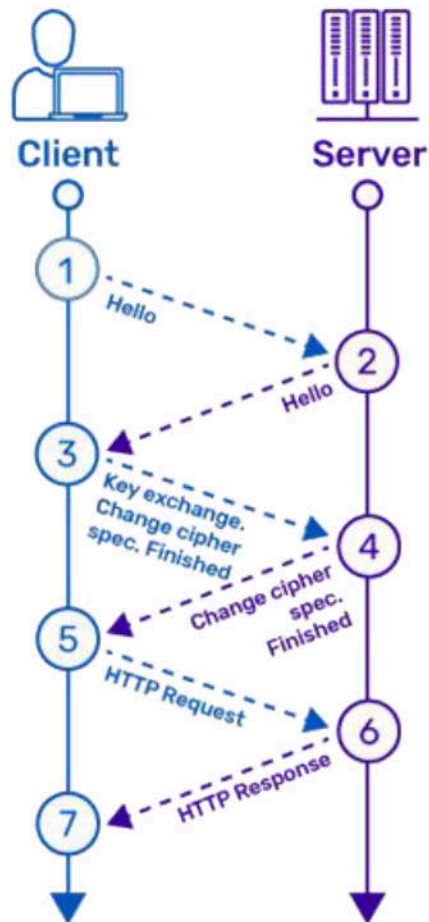
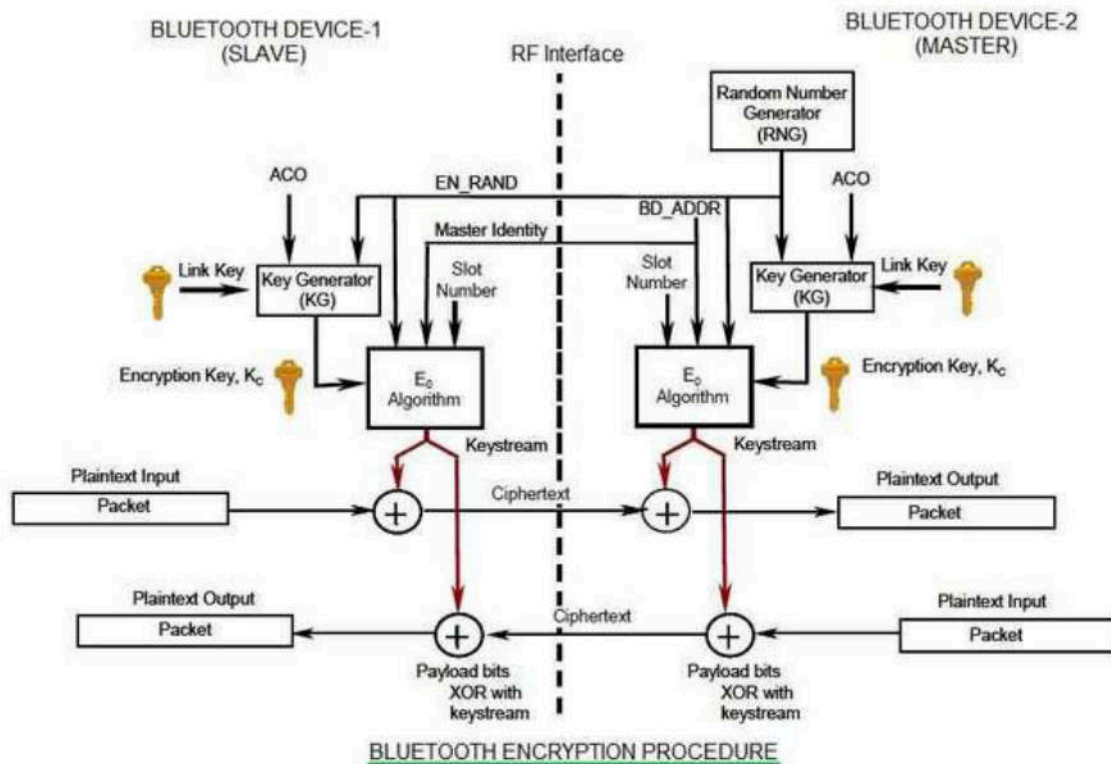
Confidentiality



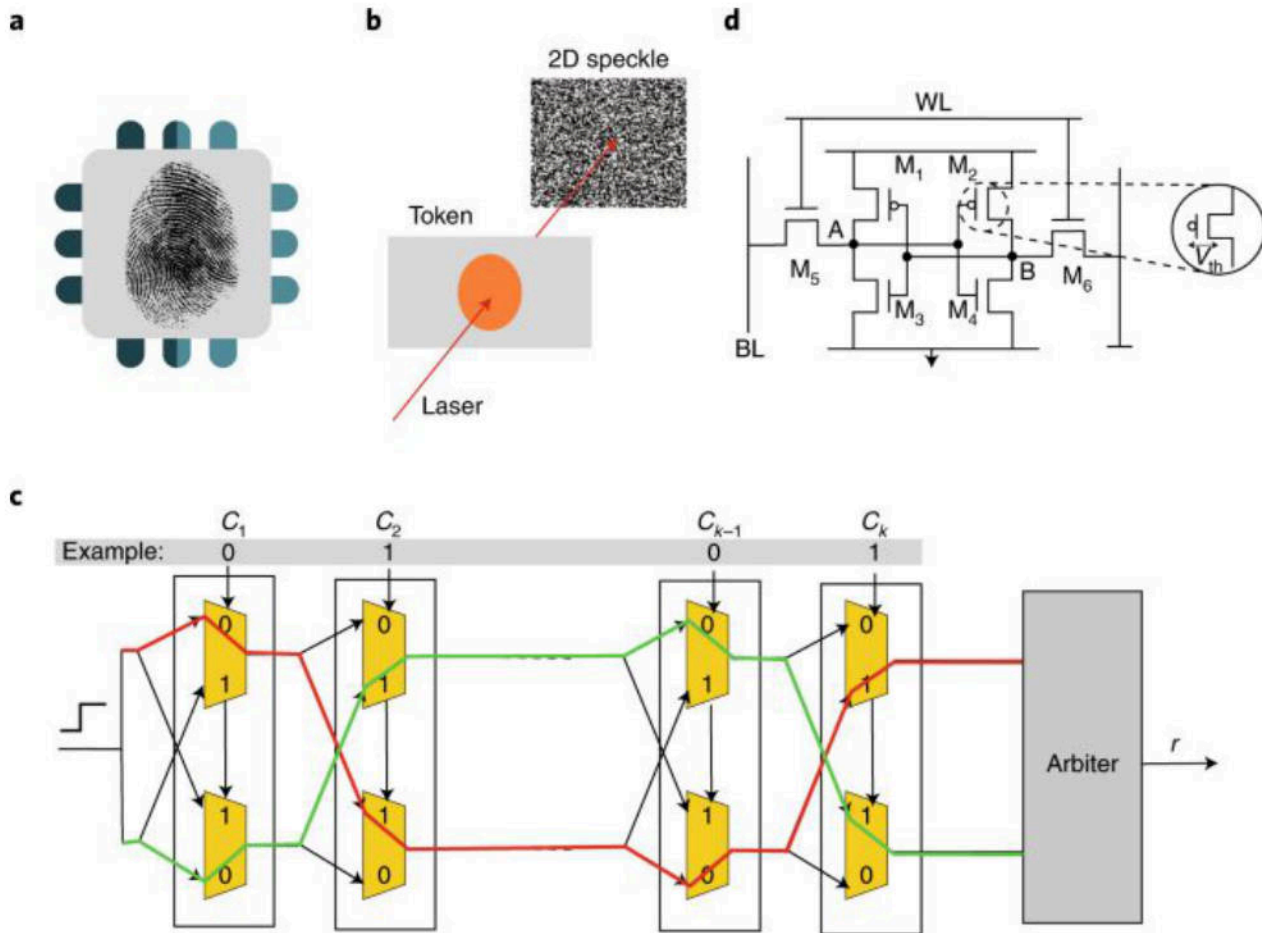
Confidentiality



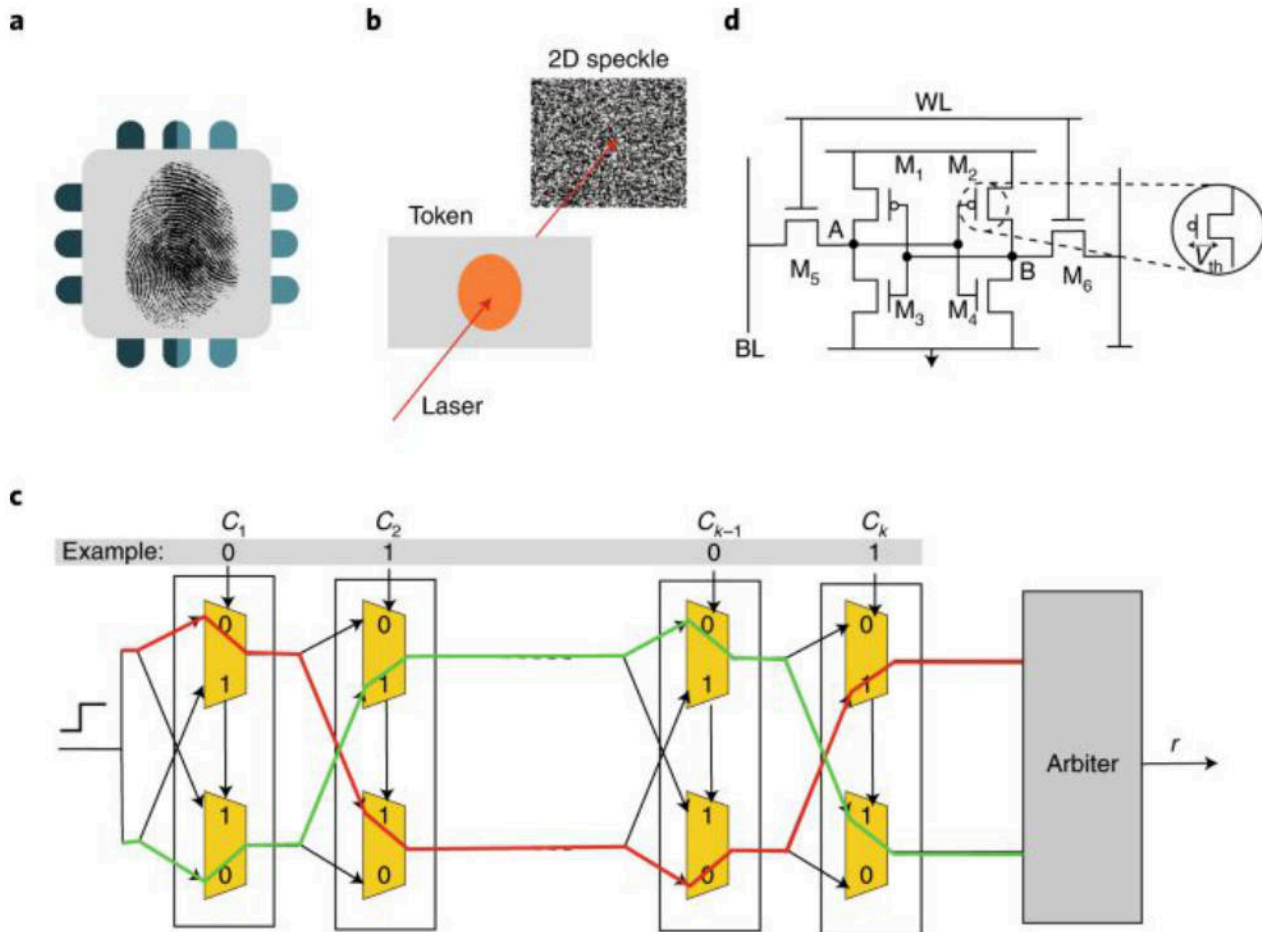
Confidentiality



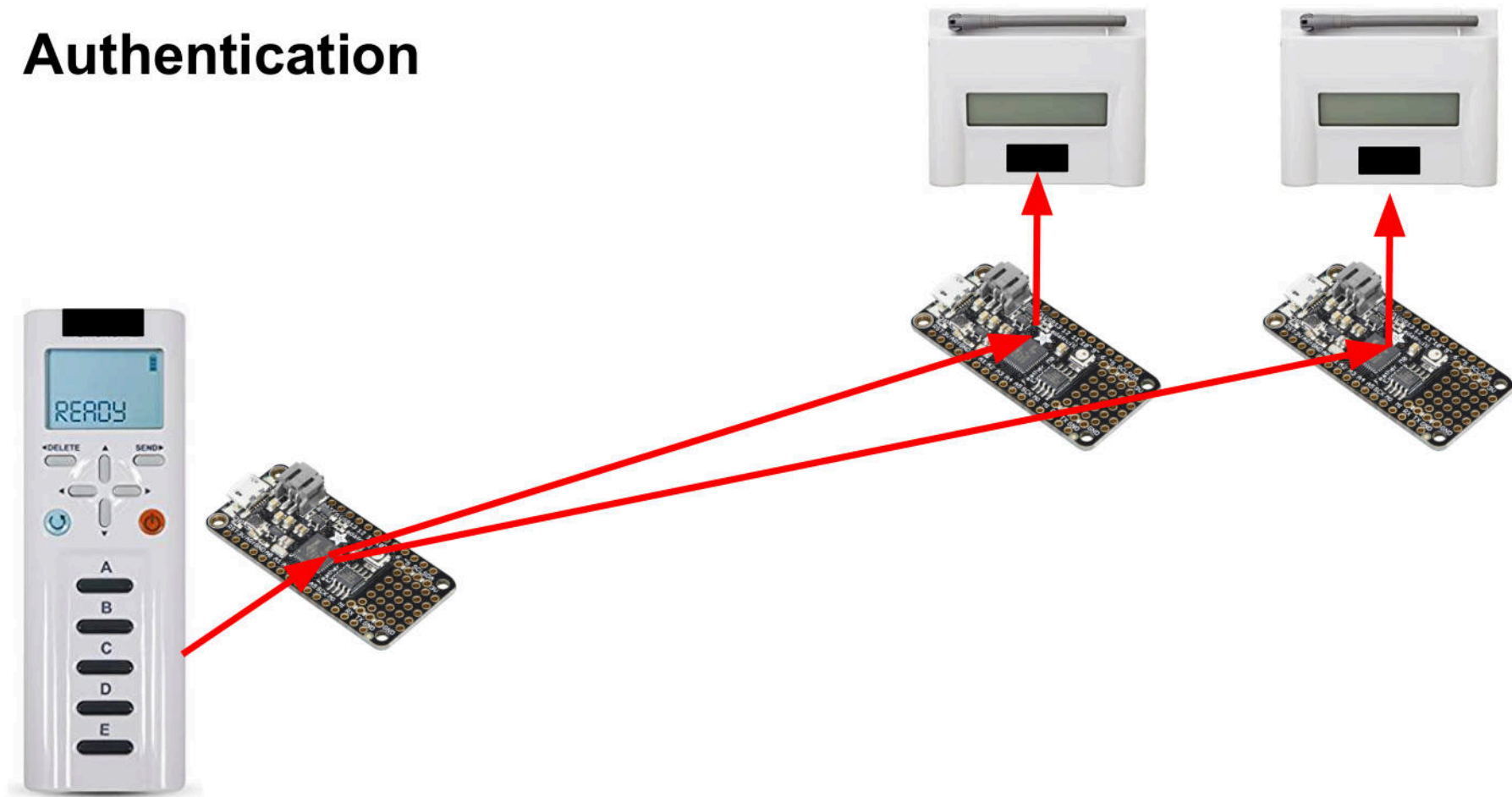
Integrity



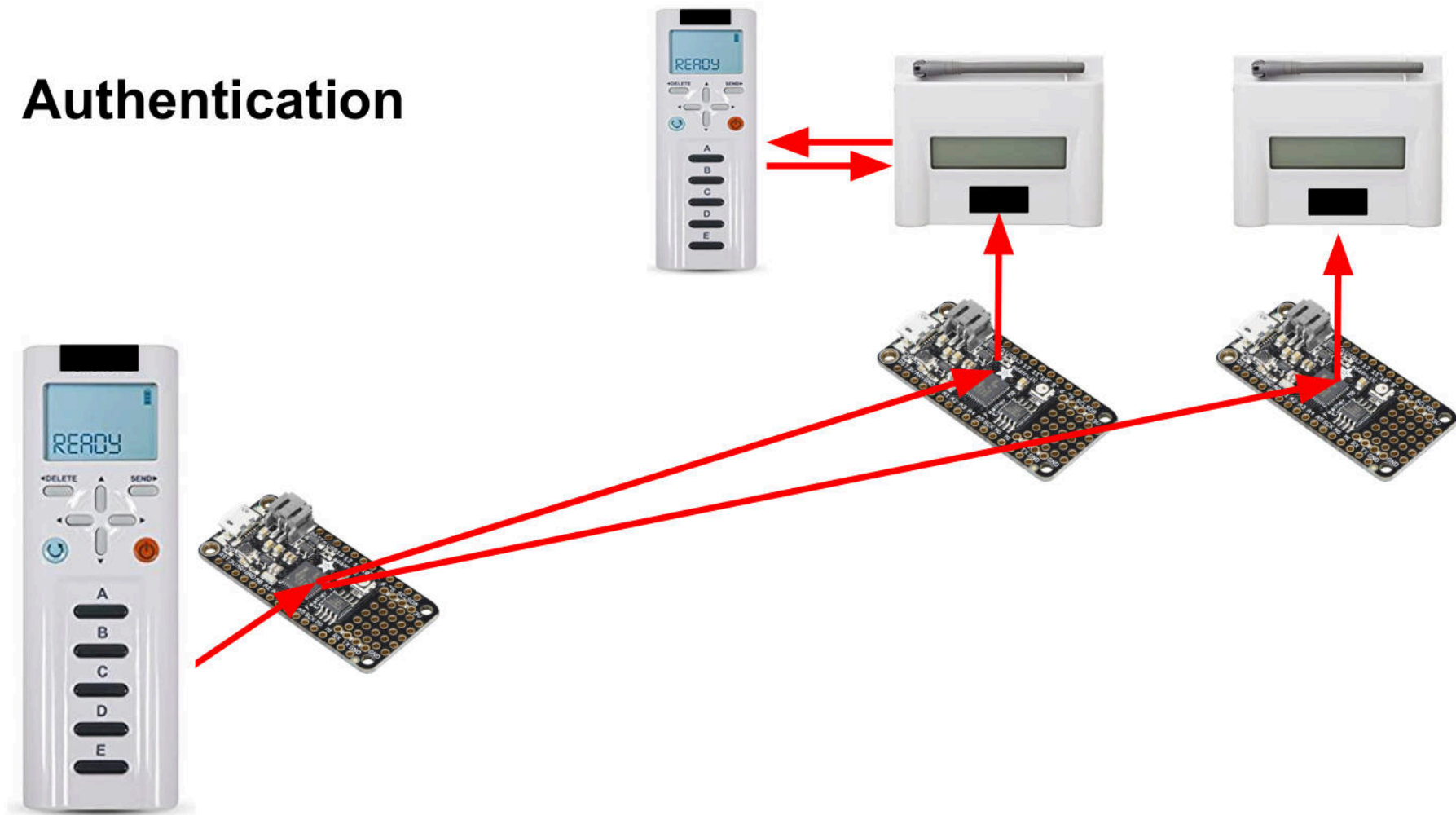
Integrity



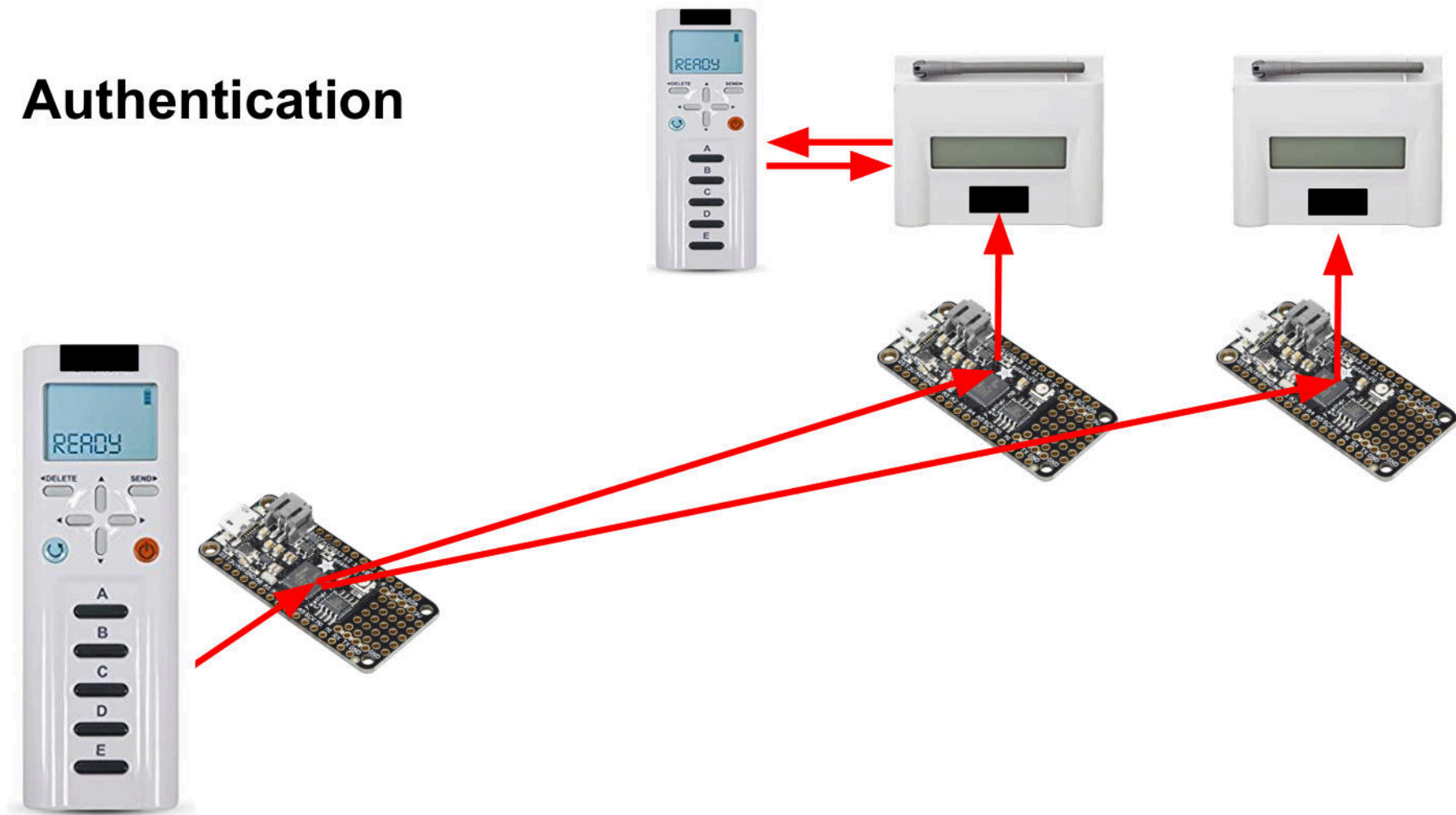
Authentication



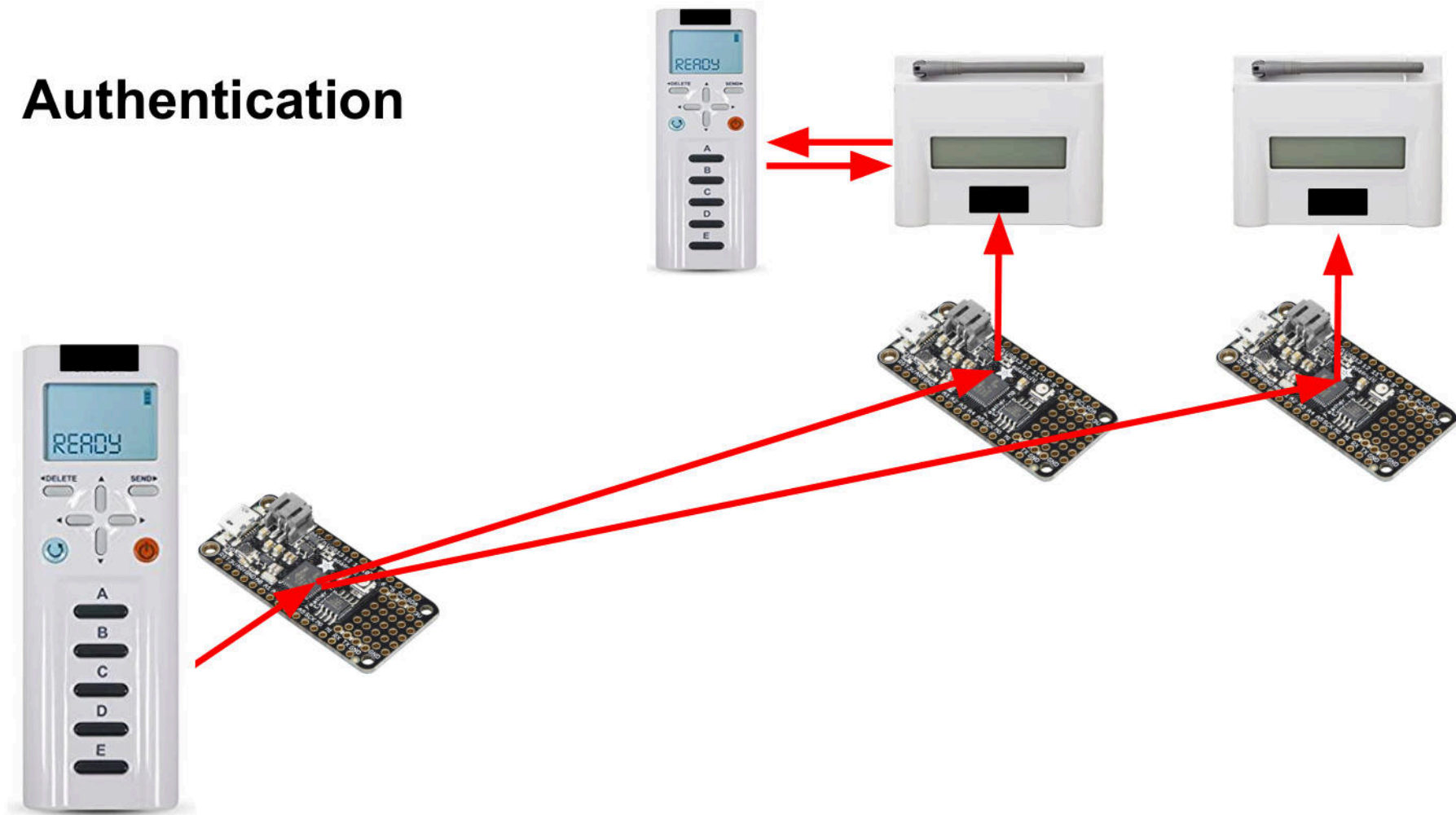
Authentication



Authentication



Authentication



Security Summary

Security Summary

Security Summary

1. Use FHSS to avoid DoS attacks
2. Use encryption in transit

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Bluetooth (or BLE)
(Kerckhoffs's Principle)

Security Summary

1. Use FHSS to avoid DoS attacks
2. Use encryption in transit
3. Use PUF
4. Use timed challenge-response



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to me ▼

Mar 14, 2020, 9:56 AM



Greetings from [REDACTED]

As you know, the university recommends that beginning March 13, all in-person meetings be moved to an alternative solution. Our classes will be held remotely as of March 23, 2020. Below you will find an outline of our plans moving forward.

[REDACTED] Online Strategy

Online Components: Prelecture, Checkpoint and Homework will continue to be delivered using [https://\[REDACTED\]](https://[REDACTED]) without any changes.

Lectures: Lectures will be built using your responses to the checkpoint questions, recorded and available by 2:00 pm US Central time on the day of the lecture via a link on the course schedule [https://\[REDACTED\]](https://[REDACTED]) It is expected that all students will review the lecture and attempt the interactive questions therefore all will be awarded full participation and bonus points.

Discussion: Your discussion TA will send you a zoom link which will give you access to a virtual discussion room during your usual discussion time.

Discussion Quiz: A link to a pdf of the quiz will be posted on the course syllabus on Thursday at 6 PM US Central time. You will have 24 hours to complete the quiz, and upload a pdf version of your hand written solutions to this link [https://\[REDACTED\]](https://[REDACTED])

[REDACTED] We have made a demo assignment that will allow you to upload a pdf right now just to test that the system works for you. Note, that you can upload as many times as you want before the deadline. The most recent submission will be grades. We are providing 24 hours to complete the quiz in order to accommodate a variety of schedules. However we expect that the work you submit is completed on your own, using only the course formula sheet and a calculator, just as we would do during class. Using any additional resources will be considered a violation of academic integrity.

Lab: Complete your prelab as usual. Take a picture and upload it as a pdf using same upload link [https://\[REDACTED\]](https://[REDACTED])

Mar 14, 2020, 9:56 AM



to me ▾

Greetings from [REDACTED]

As you know, the university recommends that beginning March 13, all in-person meetings be moved to an alternative solution. **Our classes will be held remotely** as of March 22.

"Our classes will be held remotely"

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Lab: Complete your prelab as usual. Take a picture and upload it as a pdf using same upload link [https://\[REDACTED\]](https://[REDACTED])

Zoom Meeting

Participants (3)

Participants (3)

Viewers (0)

Joanna Jiang (Host, me, participant ID: 10)

Jo Sanders

Leo Wang

Mute All Unmute All

Chat

From Me to Everyone: How is everyone? Let me know if you have any questions to answer.

From Me to Leo Wang (Private): Can you hear me? Let's keep the conversation going!

From Me to All Participants: Great job, guys!

To: All Participants

Type message here...

Zoom Meeting

Class for Zoom

Participants (3)

Chat

Class Mgmt

Teaching Tools

Syllabus

Assignment

Assessment / Quiz / Test

Polling

Link to Course Mgmt System

Share Screen

Presenter Mode

Whiteboard

Browse the Web

Play Video

Share Files

Zoom Meeting

Classroom Play Video

Participants (3)

Chat

Class Mgmt

Teaching Tools

Share Screen

Presenter Mode

Whiteboard

Browse the Web

Play Video

Share Files

0:00 / 6:04

Mute Stop Video Share Screen Record Feedback Raise Hand Leave